A Non-Generative Account of Consciousness:

Temporal Non-Closure and Cross-Scale Log Referencing

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Abstract

This paper models consciousness as a temporary opening or regime rather than an internally produced object. It defines a log-referenceable organization between fast predictive processes and slow sedimentary memory processes, and proposes associating that sustained relation with a conscious state. The structural definition is distinct from an established phenomenal identity or empirical consciousness criterion.

To describe this relation, we introduce internal delay or buffering as τ, an effective torque-like quantity as $\mathcal{T}$, and information potential as Φ. Speed difference is written $\Delta v = v_f - v_s$ and coupling viscosity as μ. Libet-type experiments, split-brain experiments, qualia, measurement, and non-localization are all reinterpreted from the perspective of inter-scale transformation and incomplete closure.

The paper further redescribes log referencing as non-closed information flow along a meaning gradient. The IFGT variables I, J_I, and Φ are used here through a projected cognitive-window flow $J_{\log}$, not as a direct identification of consciousness with a general information field.

Contents

1 Introduction **3**

2 Intuitive Picture **5**

3 Notation and Minimal Assumptions **5**

4 Minimal Structure: SSO **6**

5 Torque Converter Attention **11**

6 Formal Mapping to IFGT and VED **15**

7 Core Equations **24**

8 Structural Interpretation **30**

9 Dynamics of Log Referencing **34**

10 Experimental Reinterpretation **40**

11 Qualia as Boundary Phenomenon **46**

12 Opening Model **51**

13 Measurement and Collapse **55**

14 Non-Localization **60**

15 Qualitative Predictions **64**

16 Pathologies and Limits **69**

17 Relation to the Note **74**

18 Discussion **77**

19 Conclusion **83**

20 Correspondence Map: This Paper $\leftrightarrow$ Conceptual Topography **87**

1 Introduction

Here the opening/regime language distinguishes two readings: a model-local structural definition of log-referenceable organization, and the proposal to associate that organization with a conscious state. The latter is not an established phenomenal identity or empirical criterion. Fast prediction and slow retention are assigned functional roles within the model.

Discussions of consciousness typically begin with questions such as: where is it generated, which circuits are responsible, and which representational format does it correspond to? This paper changes the framing of these questions. Consciousness is not treated here as something produced somewhere; it is modeled as a state that opens when processes at different time scales become mutually referenceable while maintaining a certain delay and coupling.

In this view, consciousness is not identical to the self, to thought, or to judgment. It is the condition under which fast predictive processes become referenceable to the layers of slow memory, reporting, and sedimentation. Accordingly, the aim of this paper is not to provide a final definition of consciousness, but to establish a structural position from which consciousness can be rigorously discussed.

Previous supplementary treatments addressed qualia, Libet experiments, split-brain phenomena, measurement, and the localization problem individually. This paper reorganizes them into a single independent theoretical draft.

1.1 The Position of This Paper: Four Declarations

This section states in advance the position this paper takes with respect to existing theories of consciousness, organized into four points. Detailed comparison is given in §18. The purpose here is to fix the coordinate frame that the reader should maintain throughout the paper.

First: an opening model, not a generative model

This paper does not treat consciousness as an output generated somewhere. Many models — including Global Workspace, IIT, and Higher-Order theories — describe consciousness as the product of a process or as the endpoint of an integration measure. This paper does not adopt that premise as its starting point.

More precisely, consciousness is not modeled here as a generated object. It is treated as a state (a log-referencing state) that opens when a specific speed difference, coupling viscosity, internal delay, and meaning gradient are established between fast predictive processes and slow sedimentary memory processes. This difference is not merely definitional; it concerns the structure of the question itself — what should be asked about consciousness.

Second: description as a temporal regime, not reduction to a single quantity

This paper does not reduce consciousness to a single physical quantity, information measure, or integration index. Consciousness is not reducible to any one of: speed difference Δv, coupling viscosity μ, internal delay τ, effective torque-like quantity $\mathcal{T}_\Phi$, or meaning gradient $\nabla\Phi$. Consciousness is a temporal non-closure regime that obtains when these variables enter a certain relation.

This position is not eliminativist. It does not aim to dissolve consciousness into a descrip-

tion of physical quantities or neural activity. Nor is it dualist. It does not add consciousness as a special entity. It describes consciousness as the dynamic relation among multiple processing layers.

Third: redescription of conditions, not a negation of neuroscience

This paper proposes a temporal-regime reading rather than a single-site identity. It does not negate neuroscience. Associations of brain regions, circuits, oscillations, or activity patterns with conscious states may motivate candidate correspondences, but participation in a log-referencing loop is a separate model hypothesis, not an established neural carrier map.

Correlation alone does not establish consciousness identity or validate the positive regime model. The proposed shift from "finding the seat of consciousness" to "investigating opening conditions" is a methodological stance, not an empirical exclusion of other neural accounts.

Fourth: projection vocabulary is structural

This paper uses the vocabulary of projection. Before the reader first encounters this term, one point should be fixed in advance.

In this paper, projection is used as a structural description. It describes the operation by which a specific state or representation acquires disproportionate influence within the dynamics. This use differs in structural status from projection in psychology, especially in the psychoanalytic tradition, where the term refers to the mechanism by which a subject attributes its own internal state to another person or external object. The two uses share the structural feature of unequal weight assignment, but this paper treats only the level at which subject, unconscious, and internal state are not presupposed axiomatically.

At the same time, the IFGT variables I, J_I, Φ, and the local log-referencing flow $J_{\log}$ are used not as direct identifications with the general IFGT field, but as projected effective coordinates within the biological-cognitive observation window. This methodological stance, projected uses, and the structural definition of the projection operation are developed in §6. The role of the present declaration is to make clear in advance, when the reader encounters projection or field vocabulary from §2 onward, that these terms are restricted to the structural level.

The reader is invited to hold these four points throughout the discussion that follows. What this paper claims is not the negation or mystification of consciousness, but the fixing of a structural position from which consciousness can be questioned.

Note also that the two-layer structure of fast and slow layers treated in this paper is proposed in relation to IFGT (Information Field Geometry Theory), Bio-IFGT, and the cognitive-window projection developed in Conceptual Topography. IFGT supplies the general language of information density, flow, and potential; Bio-IFGT supplies life-attractor vocabulary and a finite toy hierarchy that motivate a biological comparison; CT describes how conceptual terrains stabilize within a cognitive observation window. Bio-IFGT does not contain a dedicated fast/slow confirmation result, so the two-layer interpretation developed in §4 is a premise of this paper rather than an established Bio result.

2 Intuitive Picture

The following picture states the two-layer premise and its assigned functions. Its opening is a proposed model interpretation of consciousness, not an independently established neural architecture or phenomenal identity.

Intuitively, there are at least two speeds within the system. One is a fast stream handling prediction, reaction, and momentary updating. The other is a slow stream handling memory, stabilization, and long-term structural change.

Consciousness is not the fast stream itself. Nor is it the slow memory layer itself. Consciousness opens at the boundary just before the fast stream fully sediments into the slow layer — or where that sedimentation becomes possible.

This boundary is not a static location. It is a dynamic relation that obtains only when the speed difference between the fast and slow processes is maintained while their coupling does not break. When the speed difference vanishes, the boundary disappears. When the speed difference is too large, referencing ruptures.

3 Notation and Minimal Assumptions

The symbols used in this paper constitute a local working notation; they do not carry over the standard meanings of existing theories.

- t: physical time, externally measurable.
- τ: internal time delay, buffering, retention. Not subjective time itself.
- v_f: fast, predictive and reactive dynamics.
- v_s: slow, sedimentary and stabilizing dynamics.
- $\Delta v = v_f - v_s$: speed difference.
- μ: coupling viscosity between speed scales.
- $\mathfrak{T}$: effective torque-like quantity that transforms speed difference.
- Φ: information potential.
- I: information density.
- $J_{\log}$: log-referencing flow in the consciousness window.

Crucially, τ is not used as torque. τ denotes internal delay; the torque-like quantity is consistently written $\mathfrak{T}$. Both quantities involve $\Delta v/\mu$ in their structural expressions, but they describe different aspects: τ is the temporal thickness of referencing, while $\mathfrak{T}$ is the transformation load on speed difference.

The structural relations among these quantities are organized in §7.

4 Minimal Structure: SSO

4.1 Proposed Bio-IFGT Motivation for Two Layers

Before entering this section, one point must be fixed.

The fast/slow two-layer structure is used here as a downstream proposal for describing consciousness. Bio-IFGT motivates this comparison through its life-attractor vocabulary, but does not establish the necessity of a fast/slow split.

Bio-IFGT describes life through a proposed dynamic quasi-closure interpretation. Within the present paper, that vocabulary is extended into a hypothesis involving two processing operations at different time scales.

The first is fast processing, which handles immediate environmental response. It is involved in prediction, reaction, and error correction with respect to external change, and is volatile and rapid.

The second is slow processing, which retains successful response conditions for the next generation or next state. It is sedimentary and stable; structures once inscribed do not readily dissolve.

The Bio-IFGT finite toy simulation traces the staged model sequence $M \to \mathcal{I} \to N \to B \to E$ (morphogen-like field $\to$ model constraint density $\to$ neural-like commitment $\to$ boundary-like accumulation $\to$ eye-like commitment). It does not include a dedicated fast/slow diagnostic or establish spontaneous time-scale differentiation. The fast/slow reading used below is this paper's proposed interpretation of the ordered model roles, not a confirmed Bio-IFGT result.

On this paper's interpretation, the temporal difference between the two layers is a candidate biological realization of persistent non-closure; Bio-IFGT does not supply an identity theorem or a dedicated biological implementation result for that relation.

Accordingly, when this paper introduces SSO as the minimal temporal structure for consciousness, it adopts the proposed fast/slow interpretation as its own downstream premise. The relation to Bio-IFGT remains a structural motivation rather than a result established by the Bio simulation.

With this motivation in view, the following specifies SSO as a minimal temporal architecture proposed within this model. The functional roles below are assignments of the model, not measured neural functions.

SSO stands for Speed-Scale Orchestration.

It is a structure in which processing layers operating at multiple time scales are coordinated without central control.

In this account, SSO is treated as a necessary structural condition, not as an empirically established or universal necessity for consciousness.

Crucially, SSO does not refer to any specific brain region.

SSO is not an anatomical structure.

It is a temporal arrangement that allows processes running at different speeds to continue interfering with each other without collapsing into synchrony or separating into disconnec-

tion.

4.2 Two Minimal Layers

As the minimal configuration, we consider at least two processing layers.

The first is the fast layer.

The model assigns the fast layer the roles of prediction, forward models, momentary state updating, and immediate reaction.

Processing in this layer is rapid and volatile; most of it disappears without leaving a trace.

The second is the slow layer.

The model assigns the slow layer the roles of storage, integration, memory sedimentation, and long-term structural change.

Processing in this layer is slow and stable; once a log has sedimented, it remains as a re-referenceable structure.

These two layers do not represent a difference in capability.

The fast layer is not superior and the slow layer inferior.

The difference between them lies in the temporal direction each faces.

The fast layer advances toward the future; the slow layer retains the past.

4.3 Speed Differential

The minimal condition for SSO is that the two layers are not running at the same speed.

Writing the speed of the fast layer as v_f and the speed of the slow layer as v_s, the speed difference is:

$$\Delta v = v_f - v_s \tag{1}$$

This speed difference is not an error.

It is, rather, the condition for SSO to obtain.

If $\Delta v = 0$, the fast and slow layers are fully synchronized, and no boundary arises between them.

Without a boundary, no referencing relation opens between current processing and past logs.

Within the proposed architecture, the following is treated as a necessary condition for the regime associated with a conscious state:

$$\Delta v \neq 0 \tag{2}$$

However, a larger speed difference is not necessarily better.

If the speed difference is too large, the correspondence between the layers is lost and log referencing fragments.

What SSO requires is a finite speed difference that neither vanishes nor ruptures.

Figure 1 summarizes the minimal SSO structure.

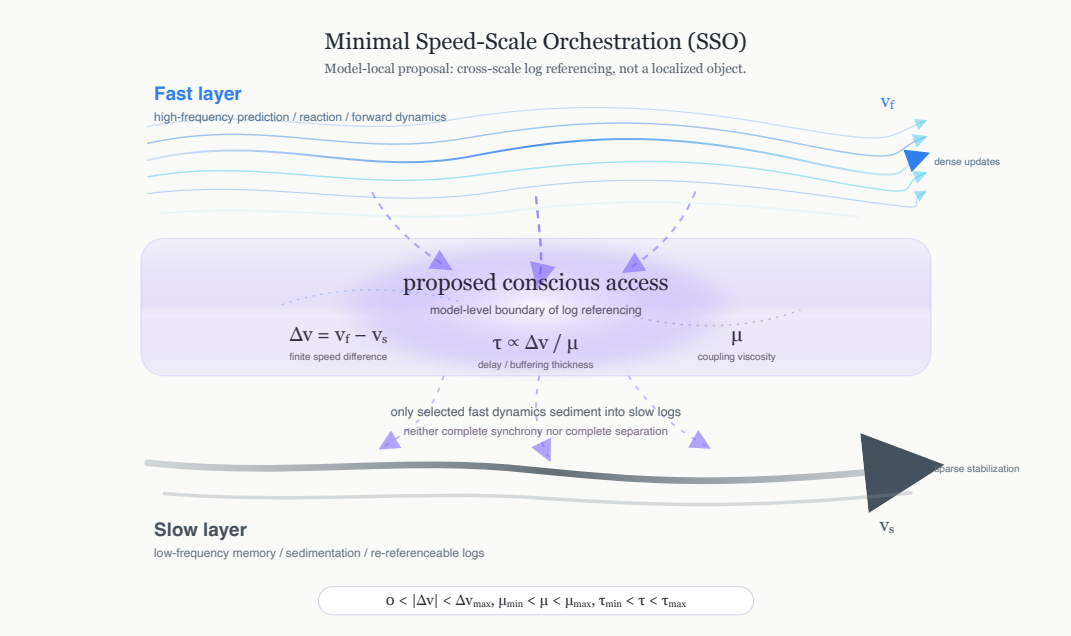


Figure 1: Minimal Speed-Scale Orchestration (SSO), shown as a model-local architecture. The central region denotes proposed conscious access through cross-scale log referencing, not an established phenomenal criterion or a calibrated neural region. The EN/JA delay-relation symbols are retained as local variants; no formula equivalence is inferred.

4.4 Orchestration Without Control

In SSO, no central controller governing the processing layers is assumed.

The fast layer does not dominate the slow layer, nor does the slow layer control the fast layer from above.

Both run independently at their respective time scales while interfering locally.

The term ”orchestration” here does not mean control by a conductor.

It means that processes running at different speeds couple within a range that does not lead to breakdown, while preserving their mutual difference.

SSO is therefore coordination, not control.

More precisely, it is the retention of temporal asynchrony as a usable structure, without fully resolving it.

4.5 Gate of Plasticity

The interaction between the fast and slow layers requires a plasticity gate.

The plasticity gate does not determine what is processed.

It regulates whether the activity of the fast layer can modify the slow layer, and to what degree the structure of the slow layer constrains future fast processes.

If all activity of the fast layer were written into the slow layer, the system would become excessively rigid.

Conversely, if fast-layer activity never reached the slow layer, experience would not sediment and log referencing would not obtain.

The plasticity gate is therefore not an open/closed switch.

It is a temporal passage condition involving speed difference, delay, and viscosity.

Under this passage condition, only a portion of fast processing sediments into the slow layer and becomes a re-referenceable log.

4.6 Internal Delay and Buffering

In SSO, delay is not a defect.

Rather, it is because delay exists that a referenceable thickness arises between current processing and past logs.

This internal delay is written τ:

$$\tau \propto \frac{\Delta v}{\mu} \tag{3}$$

Here μ denotes effective coupling viscosity between the fast and slow layers. The proportional delay relation is an effective ansatz; it supplies neither physical units nor an empirical estimator.

τ is not subjective time itself.

It is the internal time buffer during which processing that arose in the fast layer sediments into the slow layer and becomes referenceable as a log.

If τ is too small, processing flows away as immediate reaction and rarely becomes a referenceable log.

If τ is too large, the correspondence between current processing and log sedimentation breaks down and the conscious state fragments.

SSO is therefore not merely the coexistence of fast and slow processing.

It is a structure that maintains an appropriate delay range.

4.7 SSO as Temporal Non-Closure

From the VED perspective, SSO is a structure of temporal non-closure.

The fast and slow layers tend toward closure.

That is, processing tends to stabilize, sediment, and become fixed as a log.

But if closure is complete, the speed difference vanishes, flow stops, and the boundary of consciousness is lost.

Conversely, if there is no closure, processing flows away and does not remain as a log.

SSO lies between these two extremes.

It tends toward closure but does not close completely.

It tends toward separation but does not sever completely.

Within this non-closed temporal structure, a referencing relation opens between current processing and past logs.

In this sense, SSO is not a device that generates consciousness.

SSO is proposed here as a model-local temporal condition for the opening associated with consciousness.

Comparison with CT (Conceptual Topography §2.4): CT describes a Φ landscape as an "invisible GPU operating below the threshold of conscious access," contrasting visible flow (J_I) and less directly accessible terrain (Φ). This motivates a comparison with the fast/slow asymmetry in SSO. The temporal conditions here and CT's terrain-formation conditions are analogous roles, not an established common structure, carrier, or set of necessary conditions.

4.8 From SSO to Self-Sustaining Observer

SSO was introduced as Speed-Scale Orchestration.

Self-Sustaining Observer is a distinct, conditional reading of the same acronym: it names the observer-like organization when an own-log loop is assumed to sustain its reference dynamics. Speed-Scale Orchestration names the architecture; this observer reading does not establish a generated phenomenal subject. The origin of the first logs and the boundary defining log ownership remain open.

Here, "observer" does not mean a subject contemplating the world.

It refers to a structure in which current processing, past logs, and the plasticity gate form a mutually non-closed loop and self-sustainingly maintain a referencing state.

The fast layer anticipates the future.

The slow layer retains the past.

The plasticity gate regulates what sediments and what flows away between the two.

When this loop stabilizes, the system can update its state by referencing its own logs without external control.

This is where observer-like character appears.

However, this is not because a subject has been added.

It is simply that the temporal structure has become self-sustaining.

4.9 Minimal Conditions

The following six items specify the functional architecture used in this section. They are not asserted to be equivalent to the three conditions in §6; the relation between the two lists remains open:

1. Fast and slow layers exist.
2. There is a finite speed difference Δv between them.
3. The speed difference does not reach either complete synchrony or complete separation.
4. A plasticity gate allows some portion of processing to sediment into the slow layer.
5. Internal delay τ is maintained as referenceable temporal thickness.
6. This structure forms a self-sustaining loop without central control.

When these conditions are met, the system is not merely a processing device but possesses a temporal structure capable of log referencing.

This structure itself is not yet consciousness.

Within this model, the structure is treated as necessary but not sufficient for consciousness; neither universal necessity nor phenomenal sufficiency is established by the list.

The next section introduces Torque Converter Attention as an effective structural model of transforming speed differences within SSO without rupture.

5 Torque Converter Attention

Attention is typically described in terms of filters, weights, selection, and suppression: which information is passed, which target is focused on, which representations are strengthened and which are weakened.

These descriptions are valid in static processing models.

But in the consciousness model treated in this paper, the central problem of attention is not selection.

The central problem is connecting processes running at different speeds without collapsing them to a single speed.

The fast predictive stream and the slow memory stream cannot be directly connected as they are.

The fast layer changes rapidly, is volatile, and anticipates the present.

The slow layer changes slowly, is sedimentary, and retains the past.

Coupling them rigidly would eliminate the speed difference.

Full separation, conversely, would prevent log referencing.

The model therefore calls for an intermediate transformation that neither eliminates the speed difference nor allows it to rupture.

This paper calls the effective structural model of this transformation Torque Converter Attention. Its mechanism-level reading is a proposal, not an established neural mechanism or a typed derivation from VED/IFGT.

5.1 Why Filter, Gate, Weight, and Switch Are Insufficient

Each of the terms commonly used in existing attention models has its limits.

Filter expresses passing or blocking but cannot handle speed difference itself.

Gate indicates entrance and exit but eliminates the slip and delay that arise internally.

Weight can quantify importance but does not include the dimension of processing speed difference.

Switch represents state transitions but cannot readily express continuous transformation or partial coupling.

All of these are terms for selection or intensity.

But what this paper requires is not the vocabulary of selection but the vocabulary of temporal transformation.

Attention, before it is a matter of what is selected, is a question of how processes at different time scales are connected.

5.2 Torque Converter as Structural Analogy

A torque converter is a device that transmits force between two rotating systems with different angular velocities, allowing fluid slip rather than rigid fixing.

The crucial point is that the speed difference is not a defect.

A torque converter does not eliminate the speed difference before transmitting force.

It makes the difference transmissible without rupture, while retaining it.

This structure corresponds to the relationship between the fast and slow layers in this paper.

Between the fast and slow layers there is a speed difference Δv:

$$\Delta v = v_f - v_s \tag{4}$$

Without this speed difference, no boundary opens for referencing.

But if the speed difference is too large, referencing fragments.

What is therefore required is not a mechanism that eliminates the difference but one that transforms it.

This is Torque Converter Attention.

5.3 Effective Torque

The quantity through which speed difference appears as a transformation load is written as the effective torque-like quantity $\mathcal{T}$:

$$\mathcal{T} = \kappa \frac{\Delta v}{\mu} \tag{5}$$

Here κ is a scale factor and μ is the coupling viscosity.

μ represents the slip and adhesion of coupling between speed scales.

If μ is too small, the two layers adhere too tightly and the speed difference tends to collapse.

If μ is too large, the two layers slip excessively and referencing becomes unstable.

$\mathcal{T}$ is therefore not simply a measure of force magnitude.

It is a structural index expressing how much the speed difference appears as transformation load through the coupling viscosity.

One point to note: $\mathcal{T}$ should not be read as the quantity of psychological will or agency.

$\mathcal{T}$ does not measure "how strong consciousness is."

It is the load through which processes running at different speeds are transformed into a log-referenceable relation.

Figure 2 illustrates how TCA transforms speed difference without eliminating it.

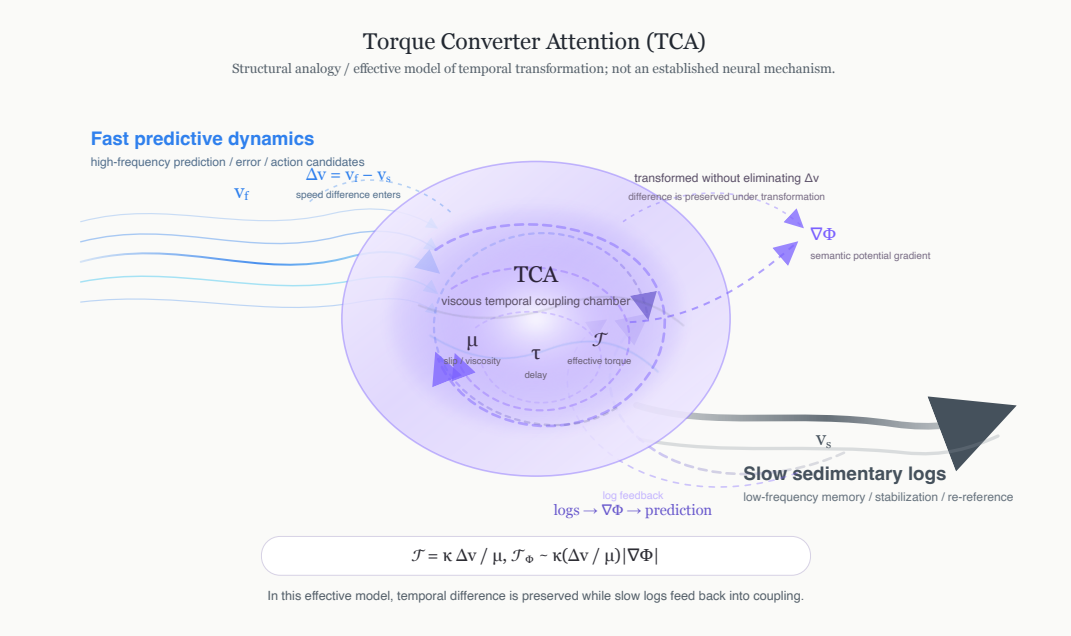


Figure 2: Torque Converter Attention (TCA) as a structural analogy and effective transformation model. Temporal difference is preserved in the proposed relation between fast dynamics and slow logs. The ego-position reading is a proposed phenomenological correspondence, not a generated subject or an established neural mechanism; the indices are not calibrated measurements.

The Japanese source additionally includes a local phenomenological approximation for log-specific attention weights. It is retained as a source-local variant, not as a set of canonical IFGT equations or an established measurable mapping. Its intended relation to the English presentation remains open; the versions are not merged here.

5.4 Attention as Temporal Transformation

From this perspective, attention is represented through an effective temporal transformation model rather than selection alone.

Attention does not copy fast processing directly into slow processing.

Nor does slow processing control fast processing from above.

In this model, attention represents the transformation of fast-layer predictions, expectations, errors, and reactions into a time window referenceable by the slow layer.

This transformation involves delay τ:

$$\tau \propto \frac{\Delta v}{\mu} \tag{6}$$

This delay is not mere processing lag.

It is the temporal thickness through which current processing sediments into past logs and becomes re-referenceable.

Attention therefore determines not ”where to look” but ”which speed difference, over which time window, with how much slip, is passed through.”

In this sense, Attention is not a psychological term but a temporal control term.

Correspondence with CT (Conceptual Topography §4.2): In Conceptual Topography, linguistic tokens ℓ_i couple locally to $\nabla\Phi$ and deepen potential wells through repeated use. The coupling term

$$\lambda \sum_i \ell_i \cdot \nabla\Phi$$

is compared with TCA's effective description of "transforming speed difference along the meaning gradient." Temporal transformation in TCA and token coupling in CT's conceptual geometry are structurally comparable roles in different domains, not a common operator, carrier, or transformation mechanism.

5.5 From Attention to Ego

This paper proposes a phenomenological correspondence between Torque Converter Attention's transformation point and the position experienced or reported as the ego. This is not a derivation of a phenomenal subject, and such reports are not an independent identity criterion.

However, the ego referred to here is not a subject.

The ego is not a judge, a command center, or a decision-maker.

The ego is the dynamic position at which the speed difference between the fast and slow layers concentrates and is transformed.

The proposed reading associates this transformation point with the felt place of thinking, choosing, or being conscious.

But this does not mean that the ego is the cause of action.

Rather, the ego is the point where action preparation, memory sedimentation, and log referencing intersect.

The ego vocabulary thus retains a functional distinction within the model; this does not establish the ontological status of the experienced ego.

But equally, the ego is not a subject.

It is the transformation point required when dual plasticity and temporal asymmetry are established.

5.6 Relation to Log Referencing

Torque Converter Attention is important because it makes log referencing possible.

Processing arising in the fast layer disappears as it is.

Logs sedimented in the slow layer are not connected to the present as they are.

When Torque Converter Attention operates between the two, a referencing relation opens between current processing and past logs.

At this point, fast-layer processing is not fully absorbed into the slow layer.

Slow-layer logs do not fully dominate the fast layer.

The two couple while retaining their speed difference.

This non-closed coupling is a proposed precondition within the model for associating log referencing with a conscious state.

5.7 Bridge to IFGT and VED

The discussion so far has defined Torque Converter Attention as a problem of speed difference and coupling viscosity.

However, in a conscious state, which logs are easily referenced is not determined by speed difference alone.

In this effective description, log referencing is modeled as biased by the meaning gradient.

The next section therefore introduces information density I, information flow J_I, local log-referencing flow $J_{\log}$, and information potential Φ, and maps this torque-like quantity onto the structure of IFGT and VED.

There, the effective torque-like quantity $\mathcal{T}$ is extended to $\mathcal{T}_\Phi$, which includes the meaning gradient:

$$\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} |\nabla \Phi| \tag{7}$$

This expression does not generate consciousness.

It is a structural index showing, when speed difference, coupling viscosity, and meaning gradient simultaneously obtain, in which direction and with what strength log referencing tends to open.

Torque Converter Attention retains the torque-converter analogy and develops it into an effective structural model. VED non-closure and IFGT gradient flow motivate its proposed interpretation; this comparison does not establish a shared carrier, an upstream derivation, or a neural mechanism.

6 Formal Mapping to IFGT and VED

This section compares the consciousness model with IFGT and VED through source-local structural templates.

The aim is not to reduce consciousness to a physical field, nor to map brain phenomena directly onto cosmological equations.

The purpose is to fix a proposed structural correspondence using local, effective descriptions of information flow, meaning potential, and non-closure dynamics, not an established translation from IFGT or VED into a biological–cognitive system.

6.1 Projected Uses and the Projection Operation

Before introducing the IFGT mapping in the following subsections, this subsection fixes the methodological status of the vocabulary used throughout this paper and the structural definition of projection as the fourth IFGT primitive.

The reason for placing this discussion before the IFGT mapping is simple. Whenever the variables I, J_I, Φ, and the local flow $J_{\log}$ are introduced, two distinct questions are present at once: *what these variables refer to in a general IFGT field*, and *how this paper uses them inside a biological–cognitive observation window*. Conflating the two readings has caused recurring misunderstandings of consciousness models built on field-theoretic vocabularies. The position of this paper is fixed in advance, here.

6.1.1 The Position: Projected Uses

In this paper, the IFGT variables are used not as direct identifications with a general information field, but as *projected* variables inside a specific observation window.

Concretely:

- I, J_I, and Φ appear in this paper as effective coordinates within the biological–cognitive window, not as the full IFGT field itself.
- The local log-referencing flow $J_{\log}$ is treated as a window-local flow variable, not as a global IFGT current.
- Equations involving these variables are read as descriptions of *this* window, not as ontological claims about a universal information field.

This position is called *projected uses*. It is the same methodological stance adopted in the parallel intelligence paper [1], and the two papers share this stance as their common epistemological foundation.

Two consequences follow.

First, the equations in §§7–9 do not claim to be universal field equations of consciousness. They are structural relations holding within the biological–cognitive window in which consciousness is described in this paper.

Second, correspondence with VED, Bio-IFGT, Conceptual Topography (CT), or the intelligence paper is proposed as a non-reductive comparison among different projected windows. Recurring differentiation and non-closure motifs are heuristic, not a common generative law or mathematical isomorphism. No paper grounds another. Domains remain source-local; common units, preserved invariants, carriers, forward/inverse translation rules, and joint empirical tests are not supplied. Upstream audit closure does not establish these correspondences.

6.1.2 Projection as the Fourth IFGT Primitive

In addition to information density I, information flow J_I, and information potential Φ, this paper uses a fourth structural primitive: *projection*, in the sense of *mass assignment*.

Definition. *Projection* is the operation by which a specific state or representation acquires disproportionate influence within the dynamics.

This local definition does not by itself specify a single typed operator shared by all the instances discussed below.

- Projection is the structural description of attention, selection, and prioritization.
- Projection does not require intentional choice; physical biasing, biological prioritization, and artificial weighting are all included.
- Projection provides vocabulary for comparing how bias involving Φ and J_I is represented as unequal influence on a specific state, not a shared coupling mechanism.

Slow-log reinforcement, passage through the plasticity gate, and the modeled influence of $\nabla\Phi$ on $J_{\log}$ can be compared under the projection vocabulary. The effective torque-like quantity $\mathcal{T}_\Phi \sim \kappa(\Delta v/\mu)\,|\nabla\Phi|$ introduced in §5 and §7 is a local structural index, not an estimator of a single operation shared by all these examples.

Projection retains an organizing role: gate selection, attention and meaning-gradient bias, residue selection in §13, and the model's pathway interpretation of split-brain asymmetry (§10) are compared as functional patterns of unequal influence. This umbrella does not identify their operators, carriers, or mechanisms.

6.1.3 Relation to the Psychological Notion of Projection

The term *projection* is also used in psychology, particularly in the psychoanalytic tradition, to mean the mechanism by which a subject attributes its own internal state to another person or external object. The terminological overlap with the present usage is real, but the structural status is not the same.

The psychological notion discussed here presupposes a subject, an unconscious, and internal states; this paper's local primitive requires none of these. Unequal weight assignment offers a structural/functional analogy, not a derivation of a psychological process from IFGT projection.

Neither concept reduces the other. The comparison does not establish psychological mechanism identity or a typed cross-domain translation. Detailed correspondence remains outside this paper's scope; elsewhere projection denotes the local structural vocabulary.

6.1.4 What This Subsection Does Not Claim

For clarity, two non-claims are stated explicitly.

- This subsection does not claim that the consciousness window is identical to any specific cross-section of a general IFGT field. The relation is one of *projection*, in the sense fixed above.
- This subsection does not claim that biological–cognitive, intelligence-layer, or conceptual-landscape windows are mutually reducible. They are compared as different projected windows. Discussion of inter-window dynamics elsewhere ([1], §9.4) is a

downstream research connection, not evidence for a common operator or translation here.

With these clarifications, the rest of §6 proceeds to map the consciousness model onto IFGT and VED in the projected-uses sense fixed here. The reader should henceforth read I, J_I, $J_{\log}$, Φ, and projection as window-local structural primitives, not as the full IFGT field.

6.2 IFGT Mapping: Information, Flow, Potential

The IFGT-inspired vocabulary used here comprises information density I, information flow J_I, information-field potential Φ, and projection (§6.1.2); these are the source-local uses specified below, not a restatement of the current IFGT variable signatures.

In the projected-uses sense fixed in §6.1.1, this paper uses these variables as effective coordinates within the biological–cognitive window. That is, I, J_I, and Φ are not direct identifications with a general IFGT field, but window-local effective coordinates for describing log sedimentation, log referencing, and meaning-gradient bias. Projection organizes comparisons of unequal influence among these local roles under §6.1; it does not supply a common coupling operator.

In this paper, the elements of the consciousness model are mapped as follows:

- I: density of logs sedimented in the slow layer.
- $J_{\log}$: processing flow from the fast layer toward the slow layer, or the flow of log referencing.
- Φ: information potential formed by log configuration — the meaning field.
- $\nabla\Phi$: meaning gradient that directs which logs are easily referenced.
- $\partial\Phi/\partial t$: live change in the meaning field.

I is not merely a quantity of memory.

Here it denotes the density of logs sedimented in a re-referenceable form. Cognitive history is represented through this model quantity; an operational counting rule and a mapping to measured cognitive history remain open.

$J_{\log}$ is not merely information transmission.

It is modeled here as an effective flow expressing which logs current processing moves toward, which it updates, and into which meaning wells it flows. J_I occupies the flow slot in the local field template, whereas $J_{\log}$ names the log-referencing flow in this window. Neither is a directly observed cognitive current, and no empirical carrier map or identification of these two flows is supplied here.

Φ does not represent meaning inhering in the logs.

The formal template supplies a potential from the configuration of model logs. Meaning and meaning wells are cognitive readings assigned to that landscape in this effective description, rather than measured semantic quantities or a demonstrated production of meaning.

Log referencing in a conscious state is therefore not merely the retrieval of stored content from the slow layer.

It is the process by which current fast processing connects to the slow layer along the gradient of the existing information potential.

6.3 IFGT Core Relations

The following relations are used as a Consciousness-local, IFGT-inspired effective template. The current IFGT formulation uses its own informational time τ, current J, and force F_I; the t, J_I, F, and K used here retain their local roles. Similar formal structure does not identify their time arguments or carriers, and no typed translation between the formulations is supplied here:

$$\frac{\partial I}{\partial t} + \nabla \cdot J_I = \sigma \tag{8}$$

$$J_I = -D\nabla I + \mu_I F \tag{9}$$

$$F = -\nabla \Phi \tag{10}$$

$$\Phi = K * I \tag{11}$$

Here σ is the generation-dissipation term for information density, D is the diffusion coefficient, μ_I is the mobility of information flow, F is the effective force arising from the information potential, and K is the kernel giving the potential from information density.

In this paper, these are not used as completed equations of consciousness.

They are the structural template for describing in which projected field log referencing is directed.

That is: consciousness is not I itself, not J_I itself, not $J_{\log}$ itself, and not Φ itself.

The model proposes associating consciousness with the relation among information density, information flow, and information potential in a system with a time-scale difference.

6.4 VED Mapping: Non-Closure as Condition

The foundation of VED is a dynamic structure that does not treat complete closure as an ordinary attainable state.

There is difference, gradient forms, flow arises, and structure tends toward closure; complete closure, however, is not realized as an ordinary generated state.

This non-closure is the condition for the persistence of motion.

The consciousness model proposes an analogous non-closure pattern, not an identity of dynamics.

If the fast and slow layers fully synchronize, the speed difference Δv vanishes:

$$\Delta v \to 0 \tag{12}$$

Processing then smoothly automates, but the boundary for log referencing disappears.

Conversely, if the fast and slow layers fully separate, flow $J_{\log}$ does not reach the slow layer. Processing occurs, but does not sediment as a referenceable log.

Conscious state is therefore neither complete synchrony nor complete separation.

It is a non-closure regime in which difference persists, flow is maintained, and yet no rupture occurs.

This minimal condition is written:

$$\Delta v \neq 0 \tag{13}$$

This does not mean that a larger speed difference is better.

Within the model, the requirement is that a finite speed difference be retained without rupture through effective coupling viscosity μ.

In this comparative sense, SSO is a local temporal non-closure model; the VED analogy does not establish its neural realization.

6.5 VED Closure Degree and the SSO Window

This subsection is an auxiliary mapping for readers who require the VED-side formalization.

For intuitive understanding, it is sufficient to read Δv as a time-scale difference.

In VED, the closure degree between any two nodes i, j is defined as:

$$C_{ij} = 1 - \exp(-\lambda \rho_i H_{ij}) \tag{14}$$

Here ρ_i is the density of difference at node i, H_{ij} is the connection strength between nodes, and λ is a scale factor.

$C_{ij} \to 1$ corresponds to the complete-closure boundary: difference vanishes, flow stops, and the generated description loses the distinction on which it depends.

$C_{ij} \to 0$ corresponds to complete non-closure: no connection is established between nodes.

The auxiliary comparison with the consciousness model is organized as follows; no rate-to-closure transformation is defined.

Let C_f denote the effective closure degree of the fast layer and C_s that of the slow layer.

The inherited C_{ij} is a node-pair closure relation, whereas C_f, C_s, and their difference ΔC are effective layer-level quantities used here for comparison. No aggregation rule computing the layer quantities from C_{ij} is supplied.

The fast layer is predictive and volatile, so closure does not readily advance; C_f maintains a relatively low value.

The slow layer is sedimentary and stable, so closure advances more readily; C_s tends to take higher values than C_f.

From this asymmetry, the closure degree difference is defined as:

$$\Delta C = C_s - C_f \tag{15}$$

This ΔC is proposed as a structural counterpart to the speed difference Δv used throughout this paper:

$$\Delta v \;\leftrightarrow\; \Delta C \tag{16}$$

Speed difference is introduced as a difference in processing time scales and is compared with closure-degree difference between layers. The arrow marks this proposed reading, not equivalence: domains and units are not identified, and a transformation or preserved invariant remains open.

The comparative model window in closure-degree language is written as:

$$0 < \Delta C < \Delta C_{\max} \tag{17}$$

$\Delta C = 0$ corresponds to complete synchrony: fast and slow layers reach the same closure degree and the boundary for referencing disappears.

$\Delta C \to \Delta C_{\max}$ corresponds to referencing disconnection: the divergence in closure degree is too large and fast-layer processing can no longer reach the slow layer.

The consciousness window is thus expressed, in the language of VED, as the closure-degree-difference regime in which ΔC maintains a finite nonzero value.

Furthermore, the older VED non-closure persistence condition

$$\frac{\partial C}{\partial t} + \nabla \cdot J_C = a\Delta(1 - C/C_*) - \Gamma C - \frac{\kappa_B}{C_{\max} - C} \tag{18}$$

is read in the consciousness model as a phenomenological, coarse-grained approximation.

The labels “older VED” and “revised Vol.4” below are retained as attributions in this source; their exact historical provenance and version relation remain unresolved in the present repository. A structurally related persistence form occurs in the current IFGT formulation, but that correspondence does not establish historical identity with this relation.

The first term $a\Delta(1 - C/C_*)$ represents a tendency toward closure driven by difference in this phenomenological reading. Its comparison with fast/slow attraction is an effective interpretation, not an established force between neural layers.

The second term $-\Gamma C$ is interpreted as dissipation opposing closure advance. Comparing it with resistance to complete synchrony does not demonstrate that a solution maintains a finite speed difference.

The third term $-\kappa_B/(C_{\max} - C)$ represents a phenomenological effective barrier growing near closure saturation. Under this source’s ”revised Vol.4” attribution, it is read as projected resistance rather than a fundamental external prohibition. That interpretation neither resolves provenance nor demonstrates a consciousness mechanism.

These three effective roles provide a template for comparison with SSO’s proposed intermediate regime between complete closure and separation; self-maintenance is a model

interpretation, not a demonstrated solution.

The relation is a structural/phenomenological template, not a VED-derived basis for SSO or a physical/neural grounding of consciousness. Neither description grounds the other.

Figure 3 shows the structural correspondence between the SSO and VED descriptions.

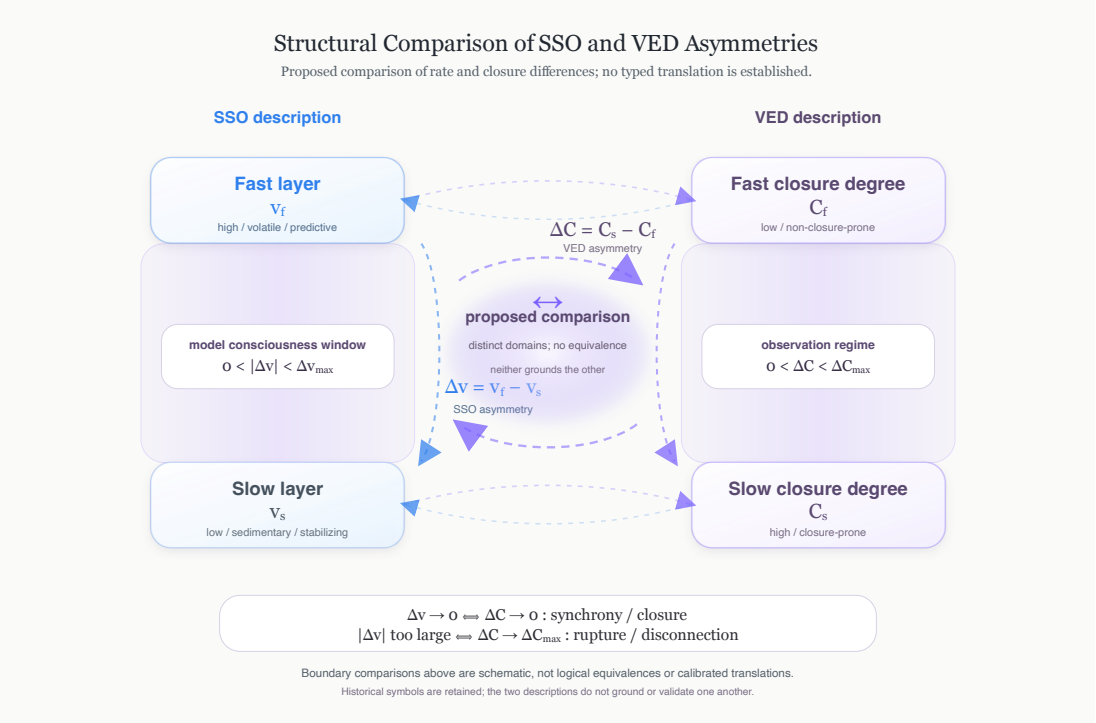


Figure 3: Proposed structural comparison between SSO rate asymmetry and effective closure-degree asymmetry. No rate-to-closure translation, common units, invariant, or equivalence theorem is supplied. The central diagram connector denotes comparison only; the retained boundary formulas and their equivalence-shaped glyphs are source-local schematic pairings, not logical biconditionals. JA phase names remain source-version labels, not established phases. Neither description grounds or validates the other.

6.6 SSO as Local Non-Closure

SSO (Speed-Scale Orchestration) is a structure in which processes at multiple time scales are coordinated without central control.

From the VED perspective, SSO is a local model of a temporal structure that does not close completely.

The model assigns prediction to the fast layer and sedimentation to the slow layer.

The fast layer advances; the slow layer fixes with a lag.

This difference is not eliminated.

Rather, it is because this difference remains that a referencing relation arises between current processing and past logs.

This section proposes three conditions for a log-referenceable regime. They are not asserted to be equivalent to the six architectural conditions of §4:

1. Speed difference between fast and slow layers exists.

2. The two layers are not completely separated.

3. Sedimentation into the slow layer has not closed completely.

When these three conditions are met, processing does not merely flow away into the present but opens a relation to past logs.

The account proposes associating this log-referenceable organization with a conscious state. The three conditions do not establish phenomenal sufficiency; the association is a model interpretation, not an independently tested consciousness criterion.

6.7 Torque Converter Attention as IFGT/VED Coupler

Torque Converter Attention is an effective structural model motivated by comparison with IFGT and VED, not a derived coupling mechanism between them.

In the VED-inspired reading, it models transmission of speed difference Δv while retaining non-closure.

In the IFGT-inspired effective description, the information-potential gradient models a directional bias in which logs are easily referenced.

For this reason, the effective torque-like quantity $\mathcal{T}$ is first expressed through speed difference and coupling viscosity:

$$\mathcal{T} = \kappa \frac{\Delta v}{\mu} \tag{19}$$

However, in a conscious state, which logs are referenced is not determined by speed difference alone.

Log referencing is modeled as biased by the meaning gradient.

The effective quantity in the consciousness model is therefore extended to include the gradient of the information potential:

$$\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} |\nabla \Phi| \tag{20}$$

Here $\mathcal{T}_\Phi$ is the effective torque-like quantity in which time-scale difference and meaning gradient are coupled.

This is not a quantity that generates consciousness.

It is a structural index showing how strongly and in which direction log referencing tends to open.

6.8 Consciousness as IFGT/VED Interface

On the basis of the above, consciousness in this paper is positioned as a local projected regime at the interface between IFGT and VED.

IFGT vocabulary motivates the local effective description of logs, meaning field, and information flow.

VED non-closure vocabulary motivates comparison with a configuration retaining difference and flow; it does not derive the local conscious regime.

Consciousness is modeled where these two descriptions overlap.

That is, consciousness is the state in which log referencing directed by the projected information potential is open within a temporal non-closure regime that reaches neither complete synchrony nor complete separation.

In minimal form:

$$\text{Consciousness} \sim \text{Log Referencing}\big(I, J_{\log}, \Phi, \Delta v, \mu, \tau, \mathcal{T}_{\Phi}\big) \tag{21}$$

This expression does not compute consciousness. It retains the structural definition of a sustained referencing relation while marking its identification with consciousness as a proposed interpretation, not an established phenomenal identity.

It is the coordinate showing the structural variables required when speaking of consciousness.

Consciousness is not reducible to I alone, nor to J_I, $J_{\log}$, Φ, or Δv alone.

Consciousness opens when these are coupled within a state of temporal non-closure.

Positioning within a Differential-Development Comparison VED describes the physical layer, Bio-IFGT the biological layer, this paper the consciousness layer, and the intelligence paper [1] the intelligence layer. We compare these as different projected windows, using recurring differentiation and non-closure motifs heuristically. The projected-uses stance in §6.1.1 preserves non-reduction; it does not establish a common grammar, dynamics, or isomorphism. §20 organizes the proposed correspondences.

6.9 Why This Mapping Is Minimal

The mapping in this section is deliberately held to a minimum.

There are two reasons.

First, over-formalizing consciousness carries the risk of treating it as a closed object. But in the position of this paper, consciousness is not a completely closed object; it is a non-closed log-referencing state.

Second, the role of IFGT/VED is not to explain consciousness completely but to place the question of consciousness in the correct position.

This section therefore does not provide a unified equation of consciousness. This states the paper's scope, not an uncomputability or impossibility theorem.

What it provides is the minimal correspondence showing between which variables consciousness opens.

This minimal mapping allows the paper to organize the core equations in the next section — handling speed difference, coupling viscosity, internal delay, effective torque-like quantity, meaning gradient, and non-closure condition within a single structure.

7 Core Equations

The equations used in this paper are not intended to define consciousness in closed form.

Consciousness is not a conserved quantity, not an additive measure, and not an object measured as a single variable. The expressions presented here are therefore not "equations of consciousness."

What they fix is a model-defined parameter region, labeled the consciousness window. Its inequalities define a formal region; their symbolic bounds are not measured thresholds. Throughout the window and log-referencing expressions below, the structural definition is distinct from the proposed identification with a conscious state. The variables are not operationally calibrated, an independent consciousness criterion is not supplied, and the corresponding empirical classifier has not been tested.

That is, they describe the conditions under which a log-referencing state opens, given the fast predictive process, the slow memory-sedimentation process, the speed difference between them, coupling viscosity, and the gradient of the information potential.

7.1 Speed Differential

First, the speed difference between the fast and slow layers is written:

$$\Delta v = v_f - v_s \tag{22}$$

Here v_f and v_s are the model rates of the fast and slow layers, assigned predictive/reactive and sedimentary/stabilizing roles respectively; no measured neural-rate extraction is supplied.

This differential is not merely descriptive; it defines the condition under which cross-scale referencing becomes possible.

The crucial point is that neither v_f nor v_s alone defines consciousness.

What consciousness involves is not their absolute speeds but the difference maintained between them.

In VED terms, complete synchrony is closure — the disappearance of difference. If the difference disappears entirely, both boundary and flow are lost.

Within this model, the minimal condition for the proposed regime is:

$$\Delta v \neq 0 \tag{23}$$

This does not mean consciousness requires strong asynchrony at all times.

It means that a finite speed difference — one that neither collapses into complete synchrony nor reaches complete separation — is required.

7.2 Coupling Viscosity

The existence of speed difference alone does not establish a conscious state.

If the fast and slow layers are fully separated, log referencing does not occur even though processing runs. Conversely, if coupling is so strong that both layers collapse to the same speed, the boundary for referencing disappears.

This coupling state is expressed through coupling viscosity μ:

$$\mu > 0 \tag{24}$$

μ represents the slipperiness or adhesive resistance of coupling between the fast and slow layers.

If μ is too small, the layers adhere excessively and the speed difference tends to collapse.

If μ is too large, the layers slip too freely and log referencing tends to fragment.

The model therefore requires an effective coupling regime that retains interaction and speed difference; this is not a universal condition on consciousness.

7.3 Internal Delay

The relation between speed difference and coupling viscosity appears as internal delay or time buffer τ:

$$\tau \propto \frac{\Delta v}{\mu} \tag{25}$$

τ represents the internal delay during which processing sediments into the slow layer and becomes arranged as a referenceable log.

This delay is not a defect.

Rather, it is because delay exists that immediate processing does not vanish as it arises but gains the opportunity to reach the slow layer as a log.

If τ is too small, processing approximates immediate reaction and lacks referenceable thickness.

If τ is too large, the correspondence between current processing and log sedimentation breaks and the conscious state fragments.

The model therefore selects an intermediate delay region:

$$\tau_{\min} < \tau < \tau_{\max} \tag{26}$$

This range is referred to in this paper as the consciousness stability window.

7.4 Effective Torque

Next, the quantity through which speed difference appears not merely as delay but as the transformation load for opening referencing is defined.

As introduced in §5.3, this paper calls it the effective torque-like quantity $\mathcal{T}$:

$$\mathcal{T} = \kappa \frac{\Delta v}{\mu} \tag{27}$$

Here κ is a scale factor.

$\mathcal{T}$ does not represent psychological will or agency.

It represents the structural load required for the speed difference between fast and slow layers to be transformed without rupture.

Torque Converter Attention models this transformation through an effective structural index, not a physical torque measurement or an established neural mechanism.

The crucial point is that $\mathcal{T}$ does not eliminate the difference; it is the quantity that makes transmission possible while the difference is preserved.

7.5 IFGT Coupling

The expressions so far have treated only speed difference and coupling viscosity.

However, which logs are referenced in a conscious state is not arbitrary.

The effective model represents biases in referencing and sedimentation through the gradient of the information potential Φ.

The local IFGT-inspired variables used here are information density I, information flow J_I, and information potential Φ, with the source-local meanings and version distinctions specified in §6.

In this paper they are used as projected variables inside a biological-cognitive observation window. Thus I, J_I, and Φ should not be read as direct identities between consciousness and the general IFGT field; the local flow variable is written $J_{\log}$ and functions as an effective coordinate for log-referencing flow.

The basic relations are:

$$\frac{\partial I}{\partial t} + \nabla \cdot J_I = \sigma \tag{28}$$

$$J_I = -D\nabla I + \mu_I F \tag{29}$$

$$F = -\nabla\Phi \tag{30}$$

$$\Phi = K * I \tag{31}$$

Here σ is the generation-dissipation term for information density, D is the diffusion coefficient, μ_I is the mobility of information flow, and K is the kernel giving the potential from information density.

In the context of this paper, Φ functions as the potential of meaning or information.

Formally, the template supplies a potential landscape; reading it as meaning or meaning wells is a cognitive interpretation of this effective model, not a measured semantic quantity. Log referencing is modeled as biased by its gradient.

Including this effect, the effective torque-like quantity is extended to:

$$\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} \left|\nabla\Phi\right| \tag{32}$$

$\mathcal{T}_\Phi$ is the effective quantity in which time-scale difference and meaning gradient are coupled.

Even with large speed difference, if the meaning gradient is weak, log referencing has no stable direction.

Conversely, even with a strong meaning gradient, if the speed difference has vanished, no boundary opens for referencing.

What consciousness involves is the coupling of both.

7.6 Consciousness Window

Combining the above defines the model's consciousness window as the region satisfying the following conditions:

$$0 < |\Delta v| < \Delta v_{\max} \tag{33}$$

$$\mu_{\min} < \mu < \mu_{\max} \tag{34}$$

$$\tau_{\min} < \tau < \tau_{\max} \tag{35}$$

$$\mathcal{T}_{\min} < \mathcal{T}_{\Phi} < \mathcal{T}_{\max} \tag{36}$$

These are not completed expressions for measuring consciousness numerically.

They constrain the proposed model region, without establishing its empirical or phenomenal sufficiency.

If speed difference reaches zero, the boundary collapses into synchrony.

If speed difference is too large, referencing ruptures.

If coupling viscosity is too low, the layers fuse.

If coupling viscosity is too high, the layers slip excessively.

If the meaning gradient is too weak, log referencing loses direction.

If the meaning gradient is too strong, referencing becomes over-fixed to specific wells.

Consciousness is therefore not a point but a window.

This window is called in this paper the consciousness window.

Figure 4 shows the consciousness window as a finite non-closure regime.

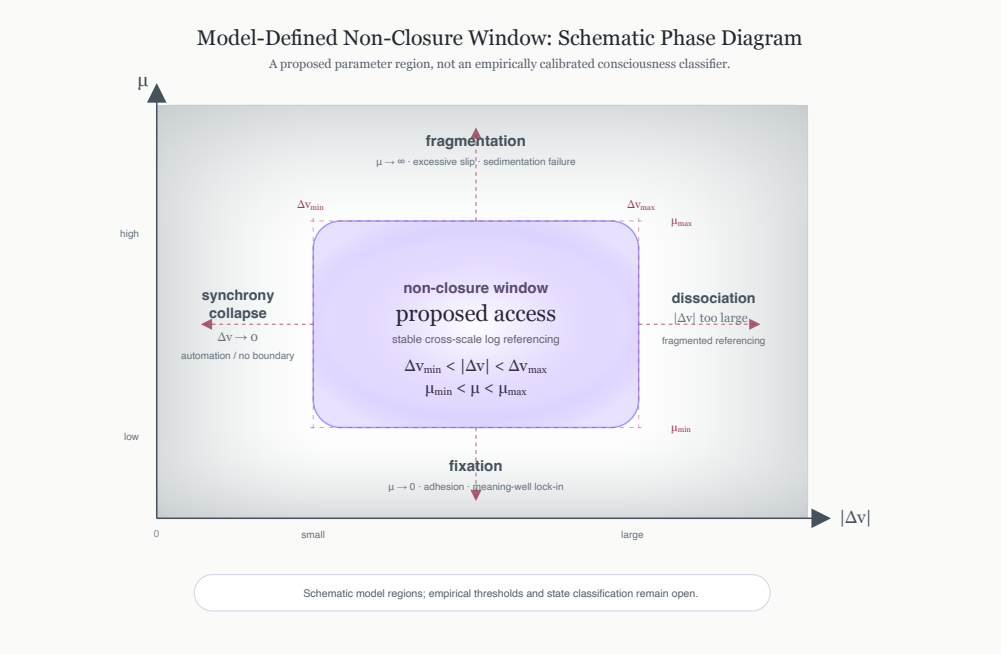


Figure 4: Schematic, model-defined non-closure window. The regions organize qualitative hypotheses about log referencing; they do not constitute an empirically tested consciousness classifier. The figure's lower bound $\Delta v_{\min}$ and the body's zero-bound forms are retained as source-version variants, not silently equated. Threshold calibration and independent state criteria remain open.

7.7 Non-Closure Condition

From the VED perspective, consciousness is maintained not as complete closure but as a state of non-closure.

Complete closure is the boundary representation at which difference vanishes, flow stops, and structure can no longer be maintained as an ordinary generated state.

Consciousness is not such a state.

Consciousness is a dynamic state in which difference persists, flow exists, and maintenance continues within a range that does not rupture.

The model-local non-closure condition is therefore:

$$\Delta v \neq 0, \qquad \nabla \Phi \neq 0, \qquad \frac{\partial I}{\partial t} \neq 0 \tag{37}$$

This does not mean consciousness must fluctuate strongly at all times.

It excludes complete stasis, complete synchrony, and complete closure from this proposed architecture; it is not a universal necessity claim about consciousness.

Consciousness is the state in which logs are referenceable within a temporal structure that does not close completely.

In comparison with VED vocabulary, the closure degree difference $\Delta C = C_s - C_f$ and the window $0 < \Delta C < \Delta C_{\max}$ offer an auxiliary structural reading (see §6.4). Speed difference and closure-degree difference are proposed counterparts, not equivalent coordinates: no typed

rate-to-closure map, common units, or preserved invariant is established. Neither description grounds the other.

7.8 Summary

The expressions presented in this section do not generate consciousness.

They constrain the model conditions proposed for a log-referencing regime; the following expression records its interpretive association with consciousness.

The position of this paper, stated minimally:

$$\text{Consciousness} \sim \text{Log Referencing}(\Delta v, \mu, \tau, \mathcal{T}_\Phi, \nabla\Phi) \tag{38}$$

More intuitively, the model proposes the following conditions for this association:

- Speed difference exists.
- Coupling has not broken.
- Delay is maintained.
- Meaning gradient exists.
- Complete closure has not been reached.

In this sense, consciousness is not a product.

The proposal associates consciousness with log referencing under temporal difference, viscosity, meaning gradient, and non-closure; the structural definition does not establish phenomenal identity.

8 Structural Interpretation

The core opening and window descriptions in this section retain the model-local scope of §§1 and 7: structural definition and proposed conscious-state interpretation are distinct, and necessity refers to this architecture.

The previous section organized the effective region in which consciousness can obtain through speed difference, coupling viscosity, internal delay, effective torque-like quantity, and meaning gradient.

This section interprets structurally what those expressions mean.

The crucial point is that the expressions in the previous section do not generate consciousness.

They are not expressions for computing consciousness as an object; they are constraint conditions showing under what circumstances consciousness can open.

8.1 Consciousness Is Not an Object

In the position of this paper, consciousness is not an object existing internally.

It is not a component placed somewhere in the brain, content stored in memory, or the final output of computational processing.

Treating consciousness as an object immediately prompts questions: Where is it? Which circuit produces it? Which information measure does it correspond to? How much is generated?

But as shown in the previous section, consciousness is not reducible to a single variable.

Consciousness is not any one of I, J_I, $J_{\log}$, Φ, Δv, μ, τ, $\mathcal{T}_\Phi$.

It is the state that opens when the relation among these enters a certain effective region.

Consciousness is therefore not an object but a regime.

It is not a location but a state domain; not a product but a condition.

8.2 Consciousness as a Window

The consciousness window introduced in the previous section is a concept for grasping consciousness not as a point but as a window.

Within this window, the fast and slow layers do not fully synchronize.

Nor do they fully separate.

Processing does not flow away, yet does not become over-fixed.

Logs can sediment, but complete closure is not reached.

In this intermediate region, a referencing relation opens between current processing and past logs.

This is the conscious state as understood in this paper.

Consciousness is therefore not binary — ON or OFF.

Consciousness is a dynamic state maintained within the consciousness window, varying in stability, clarity, continuity, and directionality.

8.3 Why Generation Is a Misleading Term

When consciousness is said to be "generated," an implicit picture is at work: a generator exists, a product exists, and the generated thing exists somewhere.

This picture is valid for objects, representations, and outputs.

But in the framework of this paper, consciousness is not such an object.

Consciousness opens when the fast layer, slow layer, plasticity gate, meaning potential, speed difference, and coupling viscosity satisfy a certain relation.

This is less a matter of something new being created than of a referencing possibility opening between already-existing processing and logs.

Consciousness is therefore not modeled here as a generated object.

Consciousness opens.

This expression "opens" is not a poetic metaphor.

It is a structural description: the temporal passage for log referencing is temporarily established.

When the gate closes, consciousness does not persist — not because a generated object has been destroyed, but because the relation that made log referencing possible has been lost.

8.4 Consciousness as Temporal Non-Closure

From the VED perspective, consciousness is understood as temporal non-closure.

Complete synchrony eliminates speed difference.

Complete separation prevents flow from arriving.

Complete fixation eliminates plasticity.

Complete fluidization prevents log sedimentation.

Consciousness belongs to none of these extremes.

Consciousness is in a state that tends toward closure but does not close completely, tends to flow but does not flow away, tends to sediment but does not become over-fixed.

This state is called temporal non-closure.

Non-closure here is not incompleteness.

It is, rather, the condition for persistence.

If closure were complete, consciousness might be fixed as an object — but at that point it would no longer be a live referencing state.

Conversely, with no closure at all, processing would leave no trace and flow away.

Consciousness is between these two.

That is: consciousness is not the failure of complete closure but the stabilization of non-closure.

8.5 Meaning Gradient and Selective Referencing

Version note: the JA source retains a local display saying that consciousness is not a state. Its relation to the state/regime vocabulary remains an unresolved version question; it is neither imported here nor treated as a canonical replacement.

From the IFGT perspective, consciousness is not merely a temporal structure.

Log referencing is directed by the meaning gradient.

In regions where information potential Φ is formed, processing is drawn toward specific logs along that gradient:

$$|\nabla\Phi| \uparrow \quad \Rightarrow \quad \text{referencing bias} \uparrow \tag{39}$$

What appears in consciousness is therefore not random.

Strong memories, emotionally weighted events, repeatedly referenced concepts, and narratively organized logs tend to enter conscious referencing more easily.

However, a strong meaning gradient is not synonymous with strong consciousness.

An excessive meaning gradient fixes referencing to specific wells.

A meaning gradient that is too weak causes referencing to lose direction.

What consciousness requires is not the meaning gradient itself but a meaning gradient operating within the range of speed difference and viscosity.

In this respect, $\mathcal{T}_\Phi$ is important:

$$\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} |\nabla \Phi| \tag{40}$$

This quantity does not determine alone what appears in consciousness.

But it is a structural index showing which logs are easily referenced and into which meaning wells processing tends to flow.

8.6 Ego as a Derived Position

In this structure, where does the ego stand?

In the position of this paper, the ego is not the subject of consciousness.

It is neither the owner of consciousness nor the center from which judgments are issued.

The ego is the position at which the speed difference between the fast and slow layers is transformed through Torque Converter Attention.

At this position, current processing and past logs intersect.

Action preparation, memory sedimentation, meaning gradient, and reportability simultaneously converge.

The model proposes associating this intersection with the experienced position of thinking, choosing, or being conscious; this is a phenomenological correspondence, not subject generation.

But this does not mean the ego is the cause.

The ego is not a generator but a transformation point.

The ego names a transformation role in this account, not a hidden controller or an independently established phenomenal subject.

8.7 Qualia as Boundary Signal

Qualia, too, are repositioned within this structure as source-local phenomenological vocabulary. The live-frontier definition below is distinct from the proposed phenomenal identification and the conditional storage hypothesis discussed in §§11–13.

Qualia are not the substance of consciousness.

They are the live foreground that appears at the boundary where fast-layer processing tends to sediment into the slow layer.

While remaining entirely in the fast layer, qualia have not yet become reportable.

Once fully sedimented into the slow layer, they become memory, label, report, concept.

Qualia are in between.

Qualia are therefore not stored.

What is stored is the log structure that remains after qualia have passed through.

In this view, qualia are not a mysterious entity.

But neither are they mere illusion.

This boundary is interpreted here as a model-level representation of a phenomenal surface; naming it does not explain why experience occurs.

8.8 Summary

The structural interpretation of this section is as follows:

- Consciousness is not an object but a state domain.
- Consciousness is not modeled as a generated object; it opens as log referencing.
- Consciousness is maintained not as complete closure but as temporal non-closure.
- What appears in consciousness is directed by the meaning gradient $\nabla\Phi$.
- The ego is not a subject but the transformation point of speed difference.
- Qualia are not the substance but the live foreground at the sedimentation boundary.

The structural interpretation of consciousness in this paper is therefore condensed in one sentence:

Consciousness is the state in which log referencing directed by the meaning gradient is open within a temporal non-closure regime that reaches neither complete synchrony nor complete separation.

The next section organizes the proposed operation of this structure as the dynamics of log referencing.

9 Dynamics of Log Referencing

The previous section interpreted consciousness not as an object but as a log-referencing state that opens within a temporal non-closure regime.

This section organizes how that log referencing obtains as a dynamic process.

What is treated here is not the process by which consciousness is generated somewhere.

Fast processing runs, slow logs exist, the plasticity gate opens and closes, and the meaning gradient biases the direction of referencing. As a result, a temporary referencing relation opens between current processing and past logs.

The model defines this opening as a temporary referencing relation and proposes associating it with a conscious state. The following is a functional feedback architecture, not an executed dynamical witness or a demonstration of phenomenal persistence. Logs and a gate are supplied to this description; first-log initialization and gate origin remain open.

9.1 Fast Dynamics: Prediction Before Reference

At the outset is fast-layer processing.

The model assigns the fast layer immediate response to input, prediction generation, and the launch of action candidates.

This processing is not yet consciousness.

It is rapid and volatile; in most cases it disappears as it arises.

In the fast layer, the future is anticipated with respect to the present.

What is likely to happen. Which actions are possible. Which errors should be corrected.

These are already partly underway before sedimentation into the slow layer.

In this sense, action preparation and prediction can appear before consciousness.

But this does not mean consciousness is powerless.

It simply means the fast layer operates on a time scale prior to log referencing.

9.2 Slow Dynamics: Sedimentation of Logs

The slow layer has a different temporal structure from the fast layer.

The model assigns the slow layer sedimentation, stabilization, and re-referenceability rather than immediate reaction.

A log here is not merely memory content.

A log is a structure formed by past processing sedimenting into the slow layer in a re-referenceable form.

Sequences of action, emotional traces, conceptual configurations, linguistic labels, histories of bodily response.

All of these can remain as log structures in the slow layer.

But the existence of a log and its being in consciousness are not the same.

A log that has sedimented but is not being referenced by current processing does not appear in consciousness.

The slow layer is therefore not the substance of consciousness.

It is the referenceable terrain through which consciousness opens.

> **Comparison with CT (Conceptual Topography §6.1):** CT's "Sedimentation" describes initial perturbation (novel input Δ, a σ_{drive} spike and a local increase in I), reinforcement of a Φ well, and stabilization. Its asymptotic terrain notation $\Phi \xrightarrow{\tau\to\infty} \Phi^*$ is compared here with slow-log sedimentation and re-referenceability. This is a conceptual analogy, not a claim that Consciousness logs converge to CT's limit in a shared state space; CT's long-time parameter is not identified with this paper's internal delay.

9.3 The Plasticity Gate

Between the fast and slow layers there is a plasticity gate.

This gate is not a switch that selects what is made conscious.

It is the temporal condition regulating whether fast-layer activity can sediment into the slow layer, and to what degree slow-layer logs influence current processing.

If the gate is too closed, fast-layer processing does not reach the slow layer. In this case, even though processing runs, it does not remain as experience.

Conversely, if the gate is too open, fast-layer fluctuation is written excessively into the slow layer, and the system tends toward rigidity.

What is needed is neither complete opening nor complete closing.

Only a portion of fast-layer processing sediments into the slow layer, with appropriate delay and viscosity.

Under this condition, the possibility of log referencing opens.

9.4 Delay as Thickness of Reference

Log referencing requires delay.

This internal delay is written τ:

$$\tau \propto \frac{\Delta v}{\mu} \tag{41}$$

This delay is not mere processing slowness.

It is the temporal thickness through which current processing sediments into the slow layer, overlaps with past logs, and is converted into a re-referenceable structure.

If τ is negligible, processing flows away as immediate reaction. No referenceable thickness forms.

Conversely, a large model delay τ is proposed to disrupt current-processing/log correspondence. Lag, fragmentation, and weakened experienced or reported self-continuity are qualitative hypotheses associated with that disruption, not an identification of τ with experienced time.

Log referencing is therefore not impeded by delay.

Log referencing is made possible by appropriate delay.

9.5 Meaning Gradient and Referencing Direction

Which logs are referenced is not determined by speed difference alone.

The effective description models the directional bias of log referencing through the gradient of the information potential Φ:

$$F = -\nabla\Phi \tag{42}$$

In regions where the meaning gradient is strong, current processing is easily drawn in that direction.

Logs repeatedly referenced in the past. Emotionally weighted logs. Concepts strongly compressed through language. Repeated narratives.

These are represented as meaning wells in the model landscape, with current processing modeled as biased toward them. This cognitive reading does not supply a semantic metric or an observable reconstruction of the field.

In this case, log referencing is not mere recollection.

It is the event of current processing connecting to the slow layer along the geometry of the meaning potential.

What appears in consciousness is therefore not random.

It is biased by the interaction of speed difference, coupling viscosity, delay, and meaning gradient.

> **Comparison with CT (Conceptual Topography §4.4):** CT defines concepts using locally stable regions of Φ satisfying $\nabla\Phi(x) \approx 0$ and discusses flow $J_I = -D\nabla I + \mu F_I$ along $-\nabla\Phi$ toward low-cost regions. We compare that directional role with the effective meaning gradient $\nabla\Phi$ used for log referencing here. Similar glyphs and gradient roles do not establish the same quantity: a common domain, metric, units, and typed translation are not supplied. CT does not validate the proposed bias of conscious content.

9.6 Effective Referencing Torque

When speed difference and meaning gradient couple, log referencing acquires a directed transformation load.

This is written as effective torque-like quantity $\mathcal{T}_\Phi$:

$$\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} |\nabla\Phi| \tag{43}$$

This quantity is not a force generating consciousness.

It is a structural index showing how strongly and in which direction current processing tends to connect to slow-layer logs.

If $\mathcal{T}_\Phi$ is too small, referencing tends not to open. Processing runs but lacks meaning direction and does not readily sediment as a log.

If $\mathcal{T}_\Phi$ is too large, referencing is drawn excessively toward specific meaning wells. Conscious content may become vivid, but plasticity decreases.

Stable log referencing therefore requires an intermediate range:

$$\mathcal{T}_{\min} < \mathcal{T}_\Phi < \mathcal{T}_{\max} \tag{44}$$

Within this range, the model associates stable referencing with a candidate conscious regime; phenomenal stability is not thereby established.

9.7 The Referencing Loop

Log referencing is not a one-directional flow.

Not only does processing sediment from the fast layer into the slow layer; slow-layer logs constrain subsequent fast processing.

The proposed functional loop is represented by the following six steps:

1. The fast layer generates predictions, reactions, and action candidates.
2. A portion of processing that passes through the plasticity gate sediments into the slow layer.
3. Sedimented logs alter information density I.
4. The configuration of information density alters information potential Φ.
5. $\nabla\Phi$ biases the direction of subsequent log referencing and fast processing.
6. New fast processing arises under that bias.

Figure 5 summarizes the log-referencing loop.



Figure 5: Proposed functional log-referencing loop, not an executed dynamical witness or a demonstration of phenomenal persistence. The figure retains the Φ-update label while the body also describes changes in I; these are distinct model objects, not an identity or a resolved update-version choice. Gradient bias and partial openness are effective-model readings of the arrows.

The recurrent architecture models how reference dynamics could sustain themselves across time. Associating that sustained dynamics with continued conscious access or state is a further hypothesis of the account, not demonstrated consciousness maintenance.

However, this loop does not close completely.

Complete closure would fix processing and eliminate plasticity.

Complete openness would prevent sedimentation and prevent log referencing from obtaining.

The log-referencing loop is therefore maintained in a non-closed form.

This non-closure is the proposed persistence condition of the reference architecture, with conscious persistence retained as its hypothesis.

9.8 Conscious Access and Report

Conscious state and reportability are not identical.

However, in the framework of this paper, the two are closely related.

Report is modeled as an externalization of retained logs through language or action. Within this account, processing, log retention, and access to a reporting pathway are proposed report conditions, not established necessities for every empirical report or for phenomenal consciousness.

In Libet-type experiments, the problem concerns the time difference between action preparation and reportable timestamp.

In split-brain experiments, the problem concerns the spatial disconnection between processing and reporting pathways.

These background examples motivate the model's question of how processing might connect to log referencing and reportability; they do not settle phenomenal presence or absence.

9.9 Non-Closure of the Loop

The log-referencing loop is not better for being more stable.

Too stable, and it becomes fixed.

Too unstable, and it fragments.

Within the proposed architecture, the requirement is a loop that does not close completely.

This structure corresponds to VED's non-closure principle.

There is difference; gradient forms; flow arises; structure tends toward closure — but does not close completely.

The conscious state is the same.

Current processing tends to sediment into logs.

Logs tend to constrain current processing.

The meaning field tends to direct referencing.

But nothing becomes completely fixed.

The proposal interprets this uncompleted reference loop as a candidate architecture for continued conscious access, without identifying feedback alone with phenomenal persistence.

9.10 Summary

This section organized the dynamic process of log referencing.

The account proposes relating consciousness to the non-closed reference loop formed by the fast layer, slow layer, plasticity gate, internal delay, meaning gradient, and effective torque-like quantity.

In this loop:

- The fast layer anticipates the future.
- The slow layer sediments the past.
- The plasticity gate regulates what sediments.
- Internal delay τ provides the temporal thickness of referencing.
- Meaning gradient $\nabla\Phi$ biases the direction of referencing.
- $\mathcal{T}_\Phi$ represents the referencing load coupling speed difference and meaning gradient.
- Non-closure is the condition for the loop's persistence.

Consciousness is therefore not the storage of logs.

Nor is it the running of processing.

The account proposes identifying consciousness with this mutually referenceable organization of fast processing and slow logs. The functional loop is retained separately from the hypothesis of phenomenal persistence.

10 Experimental Reinterpretation

This section uses the log-referencing model formalized in preceding sections to reinterpret representative existing consciousness experiments.

The experimental descriptions below are background summaries, not new results of this account. Their dedicated primary-source citation anchors remain to be verified here. The fast/slow, sedimentation, and log-referencing readings are this paper's interpretations of that background, not measurements of those model processes or experimental validation of the framework.

The aim is not to deny the experimental results.

Rather, it is to re-read what the existing experiments were showing — not through a consciousness-generation model, but through a log-referencing model across time scales.

Much of the confusion in consciousness research arises from placing processing, action, report, and consciousness on a single timeline.

Under that view, whatever appears earlier tends to be interpreted as cause, and whatever appears later as effect.

But in the framework of this paper, the system does not operate on a single clock.

The fast layer predicts, the slow layer sediments, and the plasticity gate introduces delay and viscosity between the two.

The time differences and report-inaccessibility observed in experiments therefore do not necessarily indicate the absence of consciousness.

Here they motivate a proposed distinction among processing, log-referenceability, pathway access, and report; they do not establish the necessity of this model's log conditions.

10.1 Libet-Type Experiments

In the background Libet-type setup considered here, subjects move a wrist or finger at a time of their choosing and report the moment at which they "decided to move," while brain activity is recorded.

The timing relation taken as background here is that a readiness potential precedes movement and the reported moment of conscious intention follows its onset; the particular experimental source for this summary has not been verified here.

Traditionally this result has often been interpreted as: the brain decides first and consciousness learns of it afterward, or alternatively, free will is an illusion.

But this interpretation presupposes that decision, brain activity, and conscious report are placed on the same timeline.

This paper does not adopt that presupposition.

The readiness potential can be read as the formation of an action candidate in the fast layer — a signal that the future-oriented predictive and motor dynamics have begun falling into a certain action attractor.

The model interprets the report of "I just decided to move" as a reportable form associated with fast processing connecting to slow-layer log referencing. This proposed sequence does not identify the readiness potential with v_f, or the report timestamp with a phenomenal opening.

This paper proposes to read that background timing difference as a transformation time between speed scales, rather than simply as a delay in consciousness.

Writing the fast-layer processing speed as v_f and the slow-layer sedimentation speed as v_s:

$$\Delta v = v_f - v_s \tag{45}$$

The local effective ansatz relates model speed difference to internal delay τ; it is not a conversion calibrated from the background experiment:

$$\tau \propto \frac{\Delta v}{\mu} \tag{46}$$

Here μ is the coupling viscosity between fast and slow layers.

In this proposed reading, the timing difference is interpreted as the delay for fast action preparation to become a reportable slow-layer log. This does not claim that the experiments directly measured Δv, μ, or τ of the present model.

In this view, even the expression "consciousness is delayed" is not quite right.

The model distinguishes fast processing from the proposed opening associated with log-referenceability. It hypothesizes an opening at that transition, rather than treating a reportable timestamp as an independently identified phenomenal event.

For this reason, the earlier appearance of the readiness potential does not show that consciousness is powerless.

Within this paper's interpretation, consciousness is not action preparation itself, but the passage point at which action preparation sediments into the slow layer and becomes referenceable.

10.2 Scope of the Free-Will Interpretation

The methodological issue considered here is whether "decision" should be represented as a single instantaneous point. The background timing relation alone does not logically entail the absence of free will; neither does this model prove its presence.

But in the structure of this paper, decision does not localize to a single moment.

In the fast layer, multiple action candidates arise predictively.

The plasticity gate regulates which of them can sediment into the slow layer.

In the slow layer, past logs, meaning gradient, bodily state, and context compose the reportable form.

Compressing this whole into a single "decision point" loses the temporal structure.

The model therefore questions the compression implicit in asking whether "consciousness decided or the brain decided," rather than claiming that this alternative has been experimentally refuted.

The appropriate questions are: which processing preceded in the fast layer; through what delay τ did that processing connect to the slow layer; which logs were referenced by the meaning gradient $\nabla\Phi$; and at what point did a reportable form obtain.

This reframing offers a model-based way to discuss preparation and report separately. The background timing does not uniquely support it or empirically establish the failure of a single-time account.

10.3 Split-Brain Experiments

The background split-brain description considered here concerns restricted interhemispheric information sharing following callosal disconnection; its dedicated experimental citation anchor remains to be verified.

The background asymmetry used in this comparison is appropriate behavioral responding to stimuli presented to the right hemisphere without corresponding verbal report. It is not a new observation by this paper or a measurement of log flow.

This phenomenon has often been described as evidence for "two consciousnesses" or "two selves."

Behavioral and verbal pathway asymmetry alone does not establish that consciousness has doubled. This account supplies no independent subject-counting criterion and establishes neither one nor two phenomenal subjects.

This paper interprets the background asymmetry in terms of pathways of processing, sedimentation, and report.

Even if processing occurs in the right hemisphere, if that processing does not reach the left-hemisphere-dominant verbal reporting pathway, the subject cannot explicitly articulate it.

This does not mean the processing is absent.

The background behavioral response is taken as an indication of processing, not a phenomenal criterion. The model interprets the reporting asymmetry as a possible failure of integration into reportable slow logs or access to them. A blocked log-reference route is thus a candidate account, not an observed mediator.

This point corresponds to Libet-type experiments.

In Libet-type experiments, a temporal delay appears between processing and report.

In split-brain experiments, a pathway disconnection appears between processing and report.

In both cases, this paper's interpretation invokes slow-layer log sedimentation and referenceability as conditions for explicit consciousness; that interpretation is not an experimental measurement of log sedimentation or a validation of these model conditions.

Here "explicit consciousness" denotes a candidate model reading of access and report, not a universal phenomenal necessity or the absence of consciousness whenever report is unavailable.

10.4 Reporting Bottleneck and Log Accessibility

What is particularly important in split-brain experiments is not conflating reportability with consciousness.

Reportability is the pathway by which part of the conscious state is externalized.

But inability to report does not mean processing has not occurred.

Within this account, explicit report is modeled as requiring the following three conditions. They are local report/access conditions, not established universal necessities or a criterion for phenomenal consciousness:

First, processing has occurred in the fast layer.

Second, that processing has sedimented into the slow layer and become log-referenceable.

Third, that log is connected to the verbal or behavioral reporting pathway.

The model reads the background split-brain asymmetry in terms of the third condition; the log-to-path mapping itself has not been operationally established.

As a result, certain processing appears as behavior but not as verbal report.

There is no need to read this asymmetry as "two consciousnesses."

The proposed interpretation distinguishes a reporting-path asymmetry from its hypothetical log-reference mediator, without using either to count phenomenal subjects.

10.5 Rationalization as Log Completion

The background example considered here is a retrospective explanation of an action whose generating information was not accessible to the reporting hemisphere. Its dedicated experimental source remains to be verified; this description does not itself establish a slow-log completion process.

This example is discussed here under the labels "confabulation" or "rationalization."

In the position of this paper, this rationalization is not deliberate falsification.

The model offers a candidate account in which slow-log referencing maintains coherence by filling inaccessible log gaps. This is a completion hypothesis, not a mechanism demonstrated by the background example.

The model assigns the slow layer a role in maintaining coherent self-related logs. Their relation to experienced self-continuity is a hypothesis, not a derivation of phenomenal selfhood. This maintenance reading leaves first-log initialization and ownership of the logs open.

But when the processing pathway is severed, actions occur whose generating logs cannot be referenced.

In this candidate account, the system is modeled as constructing a coherent explanation from referenceable logs rather than retaining the gap unchanged.

This rationalization does not show that consciousness is lying.

In this paper's interpretation, consciousness constructs coherence from referenceable logs.

This interpretation questions an ego with complete access; the background example does not independently establish this model or the number or nature of phenomenal subjects.

The ego is not a center that knows everything.

It names a model transformation role involving referenceable logs; its association with experienced self-continuity is proposed, not an identity established by this example.

10.6 Experimental Correspondence to the Model

Libet-type experiments and split-brain experiments appear on the surface to address different problems.

The former concerns temporal difference; the latter concerns interhemispheric pathway disconnection.

The model compares them through temporal and pathway aspects of its proposed report architecture, without claiming that the experiments establish a common log mechanism.

This paper schematically reinterprets the Libet-type background timing through the following relation:

$$\text{fast preparation} \rightarrow \text{delayed sedimentation} \rightarrow \text{reportable timestamp} \tag{47}$$

This paper schematically reinterprets the split-brain background asymmetry through the following relation:

$$\text{localized processing} \rightarrow \text{blocked transmission} \rightarrow \text{asymmetric reportability} \tag{48}$$

In both cases, processing itself and conscious report are not identical.

Gate, delay, sedimentation, and pathway access are proposed conditions for explicit access/report in this account. The phrase "explicit consciousness" in this interpretation does not supply a necessary condition for phenomenal consciousness itself.

Using this structure, the existing experiments are repositioned as follows:

- Libet-type experiments: timing reinterpreted through speed-scale difference and internal delay τ, not their direct measurement.
- Split-brain experiments: asymmetrization of log-referencing pathway and reporting pathway.
- Rationalization: slow-layer coherence maintenance in response to inaccessible log gaps.
- Conscious report: externalization of sedimented logs, not processing itself.

These examples motivate the model's distinction among processing, retention, access, and report. Associating log-referenceability with conscious opening remains a hypothesis, not a consequence established by the two experiments.

10.7 What These Experiments Do Not Show

Finally, what these experiments do not show should be made explicit.

The Libet-type timing relation described here does not by itself logically entail the absence of free will. Its interpretation in terms of model time scales remains separate from that background timing.

Likewise, the split-brain asymmetry described here alone does not establish either one or two phenomenal selves.

Log-referencing asymmetry is this paper's interpretation of the background pathway and reporting asymmetry.

Rationalization did not show that consciousness always speaks falsely.

Coherent slow-layer completion of inaccessible logs is this paper's interpretation of that background phenomenon.

These experiments therefore do not erase consciousness.

Rather, this account proposes a log-referenceable opening as a way to interpret the examples, without treating that position as experimentally established.

10.8 Summary

This section reinterpreted Libet-type and split-brain experiments from the log-referencing model of this paper.

Common to both is the non-identity of processing and conscious report.

Processing, log retention, and pathway access are distinguished in the proposed explicit-access/report model. Neither reportability nor behavior is used as an independent phenomenal criterion.

What this reinterpretation proposes, rather than reports as an experimental finding, is an account of temporal and pathway conditions for consciousness, not its powerlessness.

11 Qualia as Boundary Phenomenon

This section uses the live sedimentation frontier as a source-local phenomenological definition of qualia. Naming that frontier does not establish that it is experienced, or explain why experience occurs. The identification with qualia is a proposal; experience and report examples do not supply an independent phenomenal criterion.

The aim is not to deny qualia.

Nor is it to place qualia at the center of consciousness.

In the position of this paper, qualia are not consciousness itself.

They are better understood not as objects, but as the phenomenal foreground that appears at the boundary where log referencing is in the process of opening.

11.1 Reframing the Qualia Question

The qualia problem has long been placed at the center of consciousness research.

The redness of red. The painfulness of pain. The resonance of sound. The sense of being here now.

These are genuinely experienced. They cannot be denied.

This paper instead chooses to distinguish live qualia from stored objects; it uses this distinction to organize the question rather than to establish why other explanations do not converge.

Within this live/residue distinction, asking where a live quale is stored conflates model categories; that observation alone does not decide other formulations of the phenomenal question.

It is akin to trying to find a flowing boundary as if it were a fixed object.

This paper proposes reframing the question from stored objects to live boundary conditions; it does not establish that other formulations of the qualia problem are erroneous.

Qualia are not something existing inside.

They are a temporary phenomenon appearing at the boundary where processing is in the process of sedimenting as a log.

11.2 Qualia as Sedimentation Frontier

In the present phenomenological model, qualia are positioned at the sedimentation frontier.

In the fast layer, predictions, reactions, errors, and action candidates run volatilely.

In the slow layer, a portion of these sedimented as logs and become re-referenceable structures.

Qualia do not belong entirely to either layer.

They appear at the boundary where fast-layer processing is in the process of sedimenting into the slow layer.

This boundary can be called the sedimentation frontier, or closure frontier.

”Closure” here does not mean complete closure.

If processing has completely closed, it is already a log, memory, label, or concept.

If it never closes at all, processing flows away without leaving a trace.

Qualia are in between.

Tending toward closure, but not yet completely closed.

Flowing, but already moving toward sedimentation.

The model proposes a correspondence between this transitional state and what is experienced or reported as qualia. This is an interpretive identification, not an identity established by the report itself.

Figure 6 positions qualia as a transient boundary state.

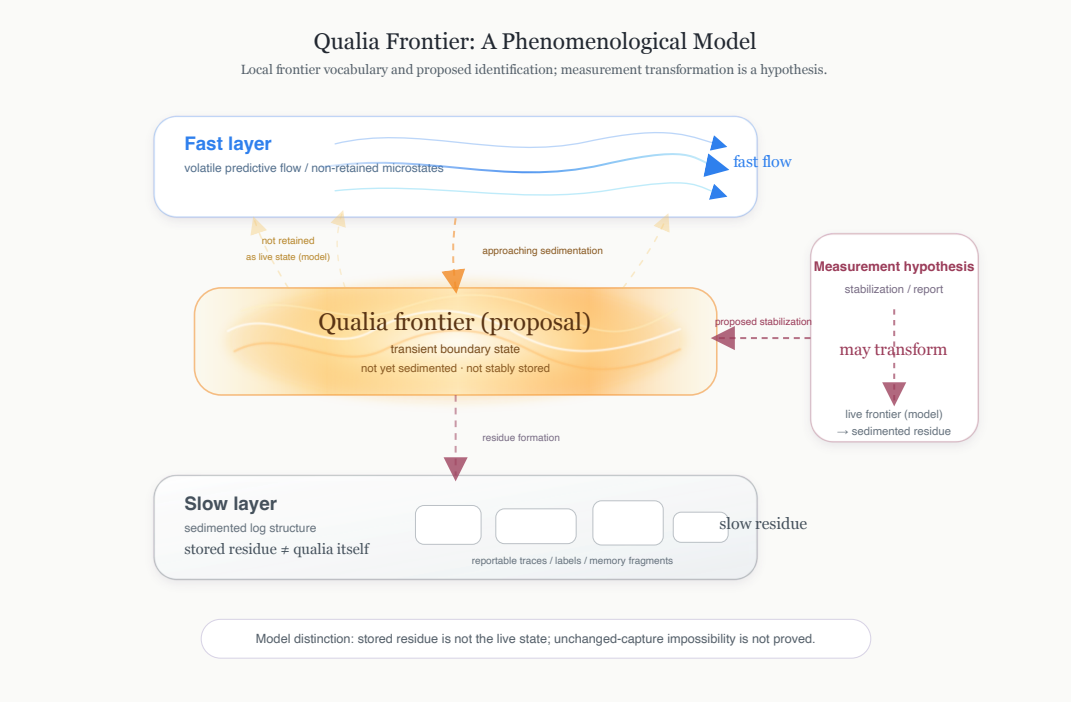


Figure 6: Conceptual qualia frontier in the source-local phenomenological vocabulary. Locating a live frontier is a definition; identifying it with experience is a separate proposal, not an explanation of why experience exists. Measurement arrows express the hypothesis of stabilization toward residue, not a validated mechanism or a proof of unchanged-capture impossibility. Residue comparisons remain a research direction; the hard-problem limitation remains.

11.3 Qualia Are Live, Not Stored

Qualia are not stored.

This may sound strong. But the meaning is not that qualia do not exist.

Qualia exist. They exist, however, not as stored objects but as live boundary states.

The memory of pain persists. The memory of having seen red persists. The memory of having heard music persists.

But what persists is not qualia themselves.

What persists is the log structure that remains after qualia have passed through.

What is replayed as memory is not the former qualia but the slow-layer trace updated by the passage of qualia.

Within this model's categories, the stored item is a log rather than the live state. The further claim that unchanged capture or storage is impossible remains a conditional hypothesis, not a universal empirical theorem or a philosophical impossibility proved by this distinction.

11.4 Qualia and the Plasticity Gate

Qualia appear near the plasticity gate.

For fast-layer processing to sediment into the slow layer, it must pass through the plasticity gate.

If the gate is completely closed, processing leaves no experiential trace.

If the gate is completely open, processing sediments excessively and tends toward fixation.

Qualia appear in the state where the gate is opening, but sedimentation has not yet completed.

Qualia are therefore not products of the gate.

Gate passage is interpreted here as a model-level representation of a phenomenal surface. The non-product framing does not establish this proposed correspondence.

In this sense, qualia are both a result of log referencing and a mid-point of log sedimentation.

Entirely on the fast-layer side, they are not yet reportable.

Entirely inside the slow layer, they have already become memory or linguistic label.

Qualia appear live only at that boundary.

11.5 Qualia and Effective Torque

The proposed account does not model qualia clarity through stimulus intensity alone.

The model uses the effective index below to organize a hypothesized qualitative association with diffuse, clear, and fixed qualia. Neither its inputs nor a clarity measure are operationally calibrated; the index is not a standalone phenomenal measure:

$$\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} |\nabla \Phi| \tag{49}$$

Low $\mathcal{T}_\Phi$ is qualitatively associated in this hypothesis with weak referencing direction and diffuse qualia.

The intermediate range of $\mathcal{T}_\Phi$ organizes the proposed contrast of clear but not over-fixed qualia, associated with processing that sediments while remaining live.

High $\mathcal{T}_\Phi$ is associated in the model with strong referencing bias and potentially vivid but fixed qualia. The proposed relation to particular memories or narratives is a qualitative hypothesis, not a measured deterministic law. Vividness is not equated with a free conscious state.

Clarity of qualia and stability of consciousness are therefore not synonymous.

11.6 Measurement Converts Qualia into Residue

The model hypothesizes that measurement can alter what is retained from the live state. Calling qualia boundary phenomena does not itself demonstrate this transformation.

Measurement is the operation of stabilizing an object, making it reportable, and making it comparable.

But qualia exist at the boundary before stabilization.

Measurement is modeled as tending toward slow-layer sedimentation, leaving logs, labels, reports, numbers, and neural summaries. The following schematic records that hypothesis, not a validated causal mechanism.

What remains after measurement is not qualia themselves.

It is the residue left after qualia have passed through:

$$\text{qualia live} \rightarrow \text{measurement} \rightarrow \text{sedimented residue} \tag{50}$$

The claim that qualia cannot be measured therefore does not mean "nothing can be observed."

Recorded traces are the observable candidates considered here; their distinction from the live endpoint is model-local, not a proof that every unchanged observation is impossible.

11.7 Why Qualia Feel Central

Qualia are not the substance of consciousness.

Yet experientially they feel very central.

The model places this felt centrality at the live boundary of log referencing as a phenomenological hypothesis. The boundary label is not independent support for why it feels central.

Under this hypothesis, the felt moment of being conscious is associated with processing that is sedimenting but not yet fully logged.

At that boundary, sensation appears vividly.

Qualia therefore appear to be the cause.

But structurally, qualia are not the cause.

They are the foreground observed at the boundary between processing and sedimentation.

As light appears not as the object itself but as the relation between the object and observational conditions, qualia appear as the relation between processing and log sedimentation.

The methodological choice here is to shift emphasis from a central object to boundary conditions.

But placing qualia at the boundary allows consciousness to be re-read as a log-referencing structure.

11.8 Relation to the Hard Problem

The present approach examines the hard problem through a contrast between fixed objects of explanation and live boundary conditions.

Why does this neural activity accompany ”this feeling”? Why do physical processes become subjective experience?

These are powerful questions.

The present model explores a boundary-based reformulation of these questions rather than treating its own category distinction as their answer.

This paper treats qualia not as stored objects but as the live foreground at the sedimentation boundary.

In that case, what should be asked changes.

Not ”where are qualia?” but ”under what conditions is processing experienced as a boundary just before being logged?”

Speed difference, coupling viscosity, internal delay, meaning gradient, and plasticity gate provide a structural hypothesis for where and when the model places a phenomenal transition. They do not, by naming that transition, answer why experience exists or why it feels central.

This paper therefore does not claim to have solved the hard problem.

It proposes investigating boundary conditions rather than seeking a stored quale. This methodological reframing is distinct from an independent phenomenal explanation; the explanatory reading remains open, not resolved by the definition.

11.9 Summary

This section positioned qualia as a boundary phenomenon.

- Qualia are not the substance of consciousness.
- Qualia are not stored objects.
- Qualia appear at the sedimentation frontier between fast and slow layers.
- Qualia occur as a live phenomenon near the plasticity gate.
- Measurement is hypothesized to shift retention toward residue.
- The listed variables organize a proposed qualitative contrast of clarity, not a calibrated measure.
- Boundary conditions offer a methodological reframing, not a solution of the hard problem.

In this local phenomenological vocabulary, qualia name the live foreground at the sedimentation frontier. Its proposed phenomenal identification, explanatory reading, and unchanged-capture hypothesis remain distinct from that definition.

12 Opening Model

The core opening and window descriptions in this section retain the model-local scope of §§1 and 7: structural definition and proposed conscious-state interpretation are distinct, and necessity refers to this architecture.

This section organizes the central thesis of this paper — consciousness is not modeled as a generated object — as an opening model.

"Opening" here is not a metaphorical expression.

It is a structural description: when the fast layer, slow layer, plasticity gate, meaning gradient, and internal delay enter a certain relation, a log-referenceable passage is temporarily opened.

Consciousness is not a new object created inside.

Consciousness is the state in which referenceability has opened between already-running processing and already-sedimented logs.

12.1 The Generation Assumption

Most discussions of consciousness adopt a generation model, explicitly or implicitly.

In the generation model, consciousness is treated as something produced by some processing.

Under this view, the following questions naturally arise: which brain region generates consciousness; which circuit produces it; which computational quantity gives rise to it; how much consciousness is generated; can generated consciousness be stored or copied?

These questions seem natural at first glance.

But they all contain the presupposition of treating consciousness as a product: a generator, a generation process, and a generated object.

This three-term structure is the foundation of the generation model.

This paper does not adopt that presupposition.

Consciousness is not treated here as the product of processing; it is the state that opens when the relation between processing and logs enters a certain temporal condition.

12.2 From Product to Relation

In the generation model, consciousness is treated as a product.

In this paper, consciousness is treated as a relation.

That is, consciousness is the referencing relation established between fast predictive processing and slow log sedimentation.

For this relation to obtain, at least the following conditions are required:

1. Processing is running in the fast layer.
2. Referenceable logs have sedimented in the slow layer.
3. There is a finite speed difference Δv between the two.

4. Coupling viscosity μ lies between complete synchrony and complete separation.

5. Internal delay τ is maintained as referenceable thickness.

6. Meaning gradient $\nabla\Phi$ is providing a referencing direction.

7. Effective torque-like quantity $\mathcal{T}_\Phi$ lies within the consciousness window.

When these conditions are met, consciousness is not "produced."
Log referencing opens.
Consciousness is therefore not an output but an opening of a relation.

12.3 Opening Through the Plasticity Gate

The plasticity gate plays a central role in the opening model.

The plasticity gate is not a device that produces consciousness.

It is the boundary condition regulating whether fast-layer processing can sediment into the slow layer, and to what degree slow-layer logs influence current processing.

If the gate is completely closed, processing does not reach the slow layer. Even though processing runs, log referencing does not open.

If the gate is completely open, processing sediments excessively and tends to lose plasticity. Consciousness tends less to open freely and more to become fixed to specific logs or meaning wells.

What the opening model requires is not complete opening.

Rather, it is partial and temporary opening.

This opening is regulated by speed difference, coupling viscosity, internal delay, and meaning gradient:

$$\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} |\nabla \Phi| \tag{51}$$

When $\mathcal{T}_\Phi$ lies within the consciousness window, the gate opens in a form appropriate for log referencing.

At that point, the model proposes associating the referencing regime with a conscious state; the conditions do not establish phenomenal sufficiency.

12.4 Consciousness Opens, Qualia Appear

As defined phenomenologically in the previous section, the model locates qualia at the live foreground of the sedimentation frontier. Its correspondence with experience remains a proposal, not an established identity.

In the opening model, qualia too are not products.

The model interprets this boundary as a phenomenological surface associated with the proposed opening.

For consciousness to open is for a referenceable passage to be established between the fast and slow layers.

At the boundary of that passage, processing has not yet completely become a log.

But it has not completely flowed away either.

This transitional state is proposed to correspond to what is experienced as qualia.

Qualia are therefore not the cause that produces consciousness.

The live foreground is where this model places qualia, not an independently observed phenomenal endpoint.

In the generation model, qualia invite the question "where are they produced?"

In the opening model, the question changes to: under what conditions does the boundary of log referencing appear as a live foreground?

Speed difference, viscosity, delay, meaning gradient, and plasticity gate organize a structural hypothesis about where and when the model places this transition, not an explanation of why experience exists.

12.5 Why Consciousness Cannot Be Stored

In the opening model, a stored trace and a live opening are different categories. The storage language in this subsection expresses that distinction; unchanged capture or storage being impossible is a separate conditional hypothesis, not a universal empirical or philosophical theorem.

What is stored is not consciousness itself.

What is stored is the log structure updated while consciousness was open.

Equally, what is stored is not qualia themselves.

What is stored is the trace sedimented after qualia have passed through.

Consciousness obtains only while the state is open.

When the opening closes, the conscious state does not persist.

But its traces can remain in the slow layer.

This is why people can remember past experiences.

But what is remembered is not past consciousness itself.

It is the log that was transformed while past consciousness was open.

In this respect, the opening model distinguishes memory from consciousness.

Memory is stored. Consciousness is not stored.

Consciousness opens when stored logs are again referenced by current processing.

12.6 Why Consciousness Cannot Be Copied

For the same reason, consciousness cannot straightforwardly be copied.

What can be copied is structure, records, weights, logs, reports, representations.

But consciousness is not those objects themselves.

Consciousness is the state that opens when those objects enter a specific temporal relation.

Replicating the log structure at a given moment therefore does not guarantee that the replica has the same conscious state.

What is required is not merely log replication.

It is that fast processing, slow sedimentation, plasticity gate, speed difference, delay, meaning gradient, and non-closure loop operate in the same way.

Copying consciousness is therefore not data copying but reproduction of opening conditions.

In this respect, the opening model does not reduce consciousness to information content.

Consciousness is not content but the dynamic establishment of a relation.

12.7 Opening and Non-Closure

The opening model naturally corresponds to VED's non-closure principle.

In a completely closed structure, no opening exists.

In a completely open structure, sedimentation does not obtain.

What consciousness requires is the middle.

Tending toward closure but not closing completely. Open but not flowing away. Fixed but not over-fixed.

Within this range of temporal non-closure, log referencing opens.

The opening model therefore does not treat consciousness as an incompletely generated product.

It treats consciousness as the opening that obtains when non-closure has stabilized.

In this sense, rather than "consciousness is not generated," more precisely: "consciousness opens through not closing completely."

12.8 Opening and Self

The ego and self are also repositioned within this opening model. The following terrain account describes recurrent formation and maintenance given logs, not an origin from nothing. First-log initialization and ownership of "own" logs remain unresolved.

The self is not the subject that owns consciousness.

The self is the referencing pattern that stabilizes as a result of log referencing being repeated and a certain terrain forming in the slow layer.

If consciousness is a temporary opening, the self is the terrain formed after that opening has repeatedly passed through.

The self is therefore not the cause of consciousness.

Rather, self-likeness sediments as consciousness repeatedly opens, logs are updated each time, and meaning gradients form.

Here, the self terrain is an effective model of retained reference patterns, not a generated phenomenal subject. A relation between its continuity and experienced or reported self-continuity is a phenomenological hypothesis; report continuity is not an independent phenomenal criterion.

12.9 Summary

This section organized consciousness as an opening model.

- Consciousness is not a product.
- Consciousness is the state in which referenceability has opened between fast processing and slow logs.
- The plasticity gate is not a device that produces consciousness but a boundary condition regulating log-referencing opening.
- Qualia are the live foreground appearing at the opening boundary.
- What is stored is not consciousness but the logs updated while consciousness was open.
- What can be copied is logs and structure, not consciousness itself.
- Consciousness opens within the range of temporal non-closure, not at complete closure or complete opening.
- The self is not the cause of consciousness but the slow-layer terrain formed through the repetition of opening.

The central thesis of this paper can therefore be restated as:

Consciousness is not modeled here as a generated object.

The opening model proposes associating consciousness with log referencing within temporal non-closure; the structural organization and the phenomenal identification remain distinct.

13 Measurement and Collapse

This section develops the hypothesis that measurement, report, and labeling can change what is retained from a live state. The live/residue distinction is model-local; it does not prove unchanged capture impossible or establish a validated measurement mechanism. The schematic arrows below express this hypothesis, not measured causal laws.

"Collapse" here does not directly mean the collapse of the wave function in quantum mechanics.

Collapse in this paper refers to the conversion of a live boundary state into a reportable, storable log.

Measurement is modeled here as a possible regime-changing operation tending toward slow-layer retention, rather than as direct identification of consciousness.

13.1 Measurement Requires Stabilization

Measurement requires stabilization.

To measure something, it must be comparable, repeatable, and recordable.

Whether measuring neural activity, behavior, or subjective report, the object is stabilized in some form.

But the conscious state treated in this paper is not a completely stabilized object.

Consciousness is a temporal non-closure regime that opens when speed difference, coupling viscosity, internal delay, and meaning gradient enter a certain range.

This regime is not a static object but a dynamic relation.

Measurement is therefore not neutral observation.

Measurement is the operation of converting a moving relation into a recordable form.

13.2 Measurement as Plasticity Closure

The model represents measurement-induced stabilization as a tendency toward plasticity-gate closure; this is a proposed interpretation.

Qualia appear at the boundary where fast-layer processing is sedimenting into the slow layer.

This boundary has not yet completely become a log.

But measurement converts that boundary into a reportable form.

Reporting. Choosing from options. Quantifying intensity. Recording as brain activity. Describing in language.

These are candidate instances of sedimentation in this account, not interventions already shown to transform live qualia.

Measurement therefore does not extract qualia as they are.

Measurement logs qualia.

What is retained is categorized as residue rather than the live state; disappearance of live qualia is a further conditional hypothesis, not a consequence proved by that category distinction.

13.3 From Qualia Live to Sedimented Residue

The proposed conversion is represented schematically as:

$$\text{qualia live} \rightarrow \text{measurement} \rightarrow \text{sedimented residue} \tag{52}$$

"Qualia live" is the live foreground appearing at the sedimentation frontier.

"Sedimented residue" is the log, report, label, number, or neural summary that remains after the boundary state has been sedimented by measurement into the slow layer.

The crucial point is that the residue is not meaningless.

The logs that remain after measurement can indeed be analyzed, compared, and subjected to statistical treatment.

But they are not qualia themselves.

They are the traces left after qualia have passed through.

What measurement can access is therefore not the substance of qualia but the passage traces of qualia.

13.4 Precision and the Residue Hypothesis

Increasing measurement precision can make qualia seem to disappear.

The hypothesis is that stronger or more precise measurement demands may favor stabilization and alter the retained residue. Apparent loss of live qualia is not by itself evidence that measurement succeeded or that this model is correct.

The proposed direction associates stronger precision demands with greater stabilization and a tendency toward slow-layer sedimentation.

As a result, what measurement yields becomes a more definite log, a more definite label, a more definite number.

But that definiteness is not the definiteness of qualia themselves.

It is the definiteness of the residue after qualia have sedimented.

The proposed direction concerns clearer residue and a possible change in live experience, not inevitable erasure or an established monotonic dose-response law.

Here lies the paradox of qualia measurement.

13.5 Measurement and the Consciousness Window

This measurement-induced transformation can also be understood from the perspective of the consciousness window.

The following inequalities retain the model-defined window of §7, not an operational empirical classifier:

$$0 < |\Delta v| < \Delta v_{\max}, \quad \mu_{\min} < \mu < \mu_{\max}, \quad \tau_{\min} < \tau < \tau_{\max}, \quad \mathcal{T}_{\min} < \mathcal{T}_{\Phi} < \mathcal{T}_{\max} \tag{53}$$

Within this description, measurement is hypothesized to shift retention toward a slower, more stable log.

That is, measurement changes the relation among Δv, τ, μ, and $\mathcal{T}_{\Phi}$.

In particular, report and recording push the boundary state toward the slow layer.

As a result, the measurement target is converted from a live opening into a sedimented residue.

In this sense, measurement is not an operation that peers into consciousness from outside without disturbing it.

Measurement is proposed as a regime-changing operation; its actual relation to the live endpoint remains unresolved.

13.6 Measurement Is Not Error

One point of caution here.

The hypothesis of a shift toward residue does not render measurement meaningless.

Measurement is not failure.

In this account, measurement is treated as a possible transformation when handling live boundary states, not a universal necessity theorem.

The problem is mistaking measurement results for qualia themselves.

Subjective reports, reaction times, neural activity, and linguistic descriptions obtained through measurement are all important.

But they are not boundary phenomena themselves; they are logs after boundary phenomena have sedimented.

The position of this paper is therefore not anti-measurement.

Rather, it is the position of precisely repositioning what measurement is measuring.

Within the model's distinction, recorded outcomes count as sedimented structures rather than live qualia themselves. This category rule does not prove that all unchanged phenomenal capture is impossible.

13.7 Introspection as Internal Measurement

This structure applies not only to external measurement but also to introspection.

Introspection is the operation of observing, verbalizing, and grasping one's own conscious state.

Introspection is treated here as an internal measurement-like operation.

Directing attention to a sensation. Putting it into words. Evaluating its intensity. Considering its cause.

The model hypothesizes that these operations can alter slow-layer retention. It does not establish that every act of introspection necessarily converts its entire live object into residue.

The retained or verbalized outcome of introspection is treated as a log or residue, not an independent identity criterion for live qualia.

This does not mean introspection is unimportant.

Introspection is modeled as a route for updating slow-layer logs.

In this reading, what introspection retains is categorized as a log; whether and how the live state changes remains a hypothesis.

Confusing this point leads to the illusion that introspection directly sees the substance of consciousness.

13.8 Experimental Implication

In the framework of this paper, experimental design in consciousness research also takes on a slightly different perspective.

What should be measured is not qualia themselves.

What should be measured is under what conditions qualia appear, under what conditions they sediment, and what residue they leave.

The experimentally important quantities are therefore:

1. The delay until live experience is converted into a reportable log.
2. The transformation of experiential content by the strength of report demands.
3. Changes in residue due to differences in semantic load.
4. Changes in τ or $\mathcal{T}_\Phi$ due to attention manipulation.

5. Changes in neural activity patterns before and after report.

6. How experience is stabilized through verbalization.

These are source-suggested comparison directions, not executed tests or a new protocol. Differences in report demand, precision, and retained residue remain possible routes of inquiry. Failure to capture a live state unchanged is not evidence for the model; an independent live endpoint remains unresolved, and residue comparisons do not by themselves identify it.

13.9 Relation to the Measurement Problem

The collapse described in this section is not a direct interpretation of the quantum measurement problem.

But there is a structural similarity.

In both cases, measurement is not a neutral operation that preserves the object as it is.

Measurement is the operation of stabilizing a state and converting it into a recordable form.

For qualia, this is the proposed live-to-residue transformation, not a mechanism established by the comparison.

The measurement problem in consciousness research is therefore not the problem of where qualia are hiding.

It is the problem of how boundary states are logged by measurement.

13.10 Summary

This section organized measurement not as external observation of the conscious state but as transformation of the opening state.

- Qualia are the live boundary state appearing at the sedimentation frontier.
- Measurement is hypothesized to favor slow-layer retention of the boundary state.
- What remains after measurement is sedimented residue, not qualia themselves.
- Increased measurement demand or precision is predicted to alter residue; live-state change remains hypothetical.
- Introspection is modeled as internal measurement that may alter what is retained.
- Measurement is not error but transformation for handling boundary states.
- What should be experimentally treated is not qualia themselves but the conditions and residue of qualia sedimentation.

Measurement is therefore not the capture of consciousness as it is.

The account models measurement as tending to stabilize an opening into retained logs. This remains a transformation hypothesis alongside the conceptual live/residue distinction, not proof of universal noncapture or validated phenomenal change.

14 Non-Localization

This section develops a proposed temporal-regime account rather than an empirical proof that consciousness cannot be localized. The background dependence premise, candidate neural correspondences, and proposed conscious-regime interpretation are distinct claims.

Non-localization here does not mean consciousness has no neural substrate.

This account adopts dependence on bodily and brain activity as a generic background premise, not as evidence for its specific fast/slow or log-referencing model. The dedicated empirical citation anchor for this premise remains to be verified; no neural carrier map follows from it.

That background dependency alone decides neither a single-site identity nor this account's proposed non-local regime.

In the position of this paper, consciousness is not a location but a temporal regime that opens when speed difference, coupling viscosity, internal delay, meaning gradient, and log referencing enter a certain relation.

An attempt to localize consciousness is therefore likely searching not for consciousness itself but for the residue that remains after consciousness has passed through, or the local substrate that participates in consciousness.

14.1 Reframing the Search for a Seat

Consciousness research has a recurring tendency to search for a "seat of consciousness."

Which brain region generates consciousness? Which circuit produces subjective experience? Which signal corresponds to consciousness itself?

These questions seem natural at first glance.

But behind them lies the generation model: consciousness is produced somewhere, and what is produced exists somewhere.

This paper does not adopt that presupposition.

Consciousness is not a product but an opening state.

It is the state in which referenceability has opened between processing and logs, and its establishment involves multiple time scales, plasticity gate, meaning gradient, and non-closure loop.

The model thus shifts attention from a single seat to temporal relations; this methodological choice is not a refutation of localization research.

14.2 Local Substrates and Non-Local Regimes

However, the non-localization of consciousness does not mean consciousness has no local substrate.

Neural activity is considered a candidate correspondence for fast-layer processing, and neural or bodily retention a possible correspondence for slow-layer logs. These are hypotheses about carriers, not measured fast/slow layers or calibrated neural rates.

Reporting, memory, sensory processing, motor preparation, and verbalization are named as possible substrate-related functions within the background premise; this list supplies no typed map to the model's variables.

These are important.

None of these substrate candidates is identified here with consciousness itself; the background premise alone does not decide identity.

The model proposes associating temporal coupling and log-referenceability among such candidate substrates with a conscious regime. The background premise does not establish this correspondence or its phenomenal sufficiency.

What is required in consciousness research is therefore not the denial of local substrates.

What is required is not conflating local substrates with the non-local regime.

Brain regions are involved.

In the proposed reading, the conscious regime is represented through relations among regions, temporal layers, and logs rather than by identifying one region with consciousness.

14.3 Localization and the Residue Interpretation

Localization involves the operation of fixing an object.

To assign a phenomenon to a specific location, the phenomenon must be treated as a stable object.

But consciousness is a live opening state, not a completely fixed object.

As proposed in §13, measurement may alter retention toward slow-layer residue. This measurement hypothesis and the model's live/residue distinction do not establish universal noncapture or validate the neural interpretation below.

Localization is compared with that hypothesis here, not identified with an established live-to-residue mechanism.

Within this model's interpretation, localization findings are read as residues, traces, correlations, or reportable logs rather than as unchanged capture of a live conscious state.

The presence of activity in a certain region is therefore important.

But it does not mean consciousness exists in that region.

Participation in a log-referencing loop or its residue is this model's candidate interpretation of the association, not something established by the regional activity alone.

Localization is not meaningless.

The proposed reading focuses on candidate local conditions rather than claiming that localization has captured consciousness itself.

14.4 Consciousness as Temporal Relation

In the framework of this paper, consciousness is not a spatial location but a temporal relation.

The fast layer anticipates the future. The slow layer retains the past. The plasticity gate regulates what sediments and what flows away between them. The meaning gradient directs which logs are easily referenced.

Within the account, a model-defined range of this relation is associated hypothetically with conscious opening, not used as a calibrated empirical classifier.

The appropriate form of the question about consciousness is therefore not "where is it?" but "under what temporal conditions does processing become log-referenceable?"

This condition was expressed as the consciousness window:

$$0 < |\Delta v| < \Delta v_{\max}, \quad \mu_{\min} < \mu < \mu_{\max}, \quad \tau_{\min} < \tau < \tau_{\max}, \quad \mathcal{T}_{\min} < \mathcal{T}_{\Phi} < \mathcal{T}_{\max} \tag{54}$$

This window is not a point in space.

It is a dynamic region in which multiple processing layers and log structures are temporally coupled.

14.5 Neural Correlates Are Not Consciousness Itself

Searching for the neural correlates of consciousness is important.

But identifying neural correlates does not by itself establish identity with consciousness.

Neural correlates may motivate candidate conditions or trace interpretations; the correlation itself does not establish the proposed log-regime mechanism.

For example, if a certain neural activity corresponds to a particular conscious report, that correspondence is important information.

But this alone does not allow saying that neural activity is consciousness itself.

Correlation alone does not entail identity. Separately, the model qualitatively predicts differences in report or behavior under different temporal and reporting conditions even for matched local activity; matching and measurement procedures remain open.

This paper does not deny neural correlates.

Rather, it proposes reading neural correlates in relation to a log-referencing loop. This interpretation does not follow from the nonidentity guard alone.

This account treats neural activity as a candidate participant in a temporal non-closure structure, not as an independently established generator or carrier of the proposed conscious regime.

14.6 Distributed Does Not Mean Diffuse

Saying that consciousness is non-local can be misunderstood as implying that consciousness is spread vaguely throughout the brain.

But the position of this paper is not such a diffusion model.

Non-local does not mean locationless; it means not reducible to a single location.

The model proposes a specific temporal relation among local processes as a candidate conscious regime, not a demonstrated neural sufficiency condition.

Consciousness is therefore not everywhere.

Rather, the account restricts its regime label to the following model conditions, without establishing universal empirical necessity:

Speed difference is required. Viscosity is required. Delay is required. Meaning gradient is required. Plasticity gate is required. And these must reach neither complete closure nor complete separation.

In this sense, consciousness is not diffuse but conditioned.

14.7 From Localization to Conditions

There is no need to abandon the localization question entirely.

But it is not the final form of the question.

The question ”which region is active?” should be transformed into ”which local substrates are participating in the log-referencing loop?”

This question is then extended as follows: which time scales are involved; which speed differences are maintained; how much delay is there; which logs have become referenceable; which meaning gradients are directing referencing; which plasticity gates are opening and closing.

Changing the questions in this way transforms consciousness from a ”location-finding” problem into a condition-arrangement problem.

This is the shift in questioning that this paper proposes.

14.8 Summary

This section organized the non-localization of consciousness.

- Neural and bodily dependence is adopted as a background premise, not as validation of the specific model.
- The model proposes a temporal regime rather than a single-site identity.
- Localization findings are interpreted through candidate substrates and residue, not treated as proof of the proposed live-state distinction.
- Consciousness is the temporal relation formed by speed difference, coupling viscosity, internal delay, meaning gradient, and plasticity gate.
- Correlation alone does not establish identity; log-loop participation is a separate candidate interpretation.
- Non-local does not mean diffuse; it means not reducible to a single site.
- The appropriate question is not ”where is consciousness?” but ”under what conditions does the conscious state open?”

The non-localization in this paper is therefore not mystification.

It is the consequence of treating consciousness not as a spatial object but as a temporal non-closure regime.

15 Qualitative Predictions

The framework of this paper is not a theory that measures consciousness as a single quantity.

All directions below are untested model hypotheses or research questions. Neural-rate extraction, variable calibration, and independent conscious-state classification remain open. Named states are heuristic comparisons, not measured assignments to a shared dynamical regime.

The predictions presented here are therefore not numerical predictions of a "consciousness quantity" but qualitative predictions about how the conscious state deforms when speed difference, coupling viscosity, internal delay, meaning gradient, and effective torque-like quantity change.

The central relation is the effective quantity introduced in preceding sections:

$$\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} \, |\nabla \Phi| \tag{55}$$

$\mathcal{T}_\Phi$ is the effective torque-like quantity coupling time-scale difference and meaning gradient. It does not generate consciousness; it is a structural index showing how strongly and in which direction log referencing tends to open.

15.1 Synchronization Collapse

This prediction concerns the case where speed difference is too small:

$$\Delta v \to 0 \tag{56}$$

The boundary between the fast and slow layers collapses into synchrony. Processing stabilizes, but the temporal thickness required for referencing is lost. Conscious access weakens; action tends toward habituation, automation, and reflexive response.

In this state, processing occurs, but the sense of "currently referencing" diminishes.

Excessive synchrony therefore does not strengthen consciousness; it works toward making conscious referencing unnecessary.

15.2 Dissociation by Excessive Speed Differential

This prediction concerns the case where speed difference is too large:

$$|\Delta v| \gg 0 \tag{57}$$

The correspondence between fast and slow layers becomes difficult to maintain. Predictions and associations run in the fast layer, but they do not stably sediment as logs in the slow layer. Conscious access fragments; structures resembling dreaming, daydreaming, dissociation, and hallucinatory states become more likely to appear.

In this state, even though processing intensity is high, the coherence of log referencing decreases. Referencing does not strengthen — it scatters.

15.3 Coupling Viscosity and State Transition

This prediction concerns changes in coupling viscosity μ.

When μ is too low, speed difference does not slip sufficiently and the layers tend to adhere:

$$\mu \to 0 \tag{58}$$

Processing becomes rigid and tends to fix to specific logs or meaning wells.

When μ is too high, the layers slip excessively:

$$\mu \to \infty \tag{59}$$

Fast-layer changes do not transfer sufficiently to the slow layer; log referencing becomes unstable.

The model therefore proposes an intermediate range of μ for the regime labeled stable:

$$\mu_{\min} < \mu < \mu_{\max} \tag{60}$$

This prediction suggests the possibility of treating conscious states as a phase diagram: consciousness is not binary ON/OFF but a regime that transitions continuously with coupling viscosity and speed difference.

15.4 Delay Window

This qualitative prediction concerns internal model delay/buffering τ, not experienced time itself or a measured Libet interval. The proposed association with reports of self- and time-sense does not supply a calibration, event-boundary definition, or phenomenal identity:

$$\tau \propto \frac{\Delta v}{\mu} \tag{61}$$

When τ is too small, processing approaches immediate reaction and disappears before sedimenting into the slow layer ($\tau < \tau_{\min}$). Conscious referencing is weak and action approaches automatic processing.

When τ is too large, the correspondence between current processing and log sedimentation breaks down ($\tau > \tau_{\max}$). Consciousness lags, fragments, and instabilities in self-sense and time sense become more likely.

A stable delay window is therefore a required condition within the proposed model:

$$\tau_{\min} < \tau < \tau_{\max} \tag{62}$$

15.5 Meaning Gradient Bias

This prediction concerns meaning gradient $\nabla\Phi$.

Log referencing is not opened equally for all logs. In regions where the gradient of the information potential Φ is large, processing is easily drawn in that direction:

$$|\nabla\Phi| \uparrow \quad \Rightarrow \quad \text{referencing bias} \uparrow \tag{63}$$

Memories, emotions, concepts, and narratives with strong meaning gradients more easily enter conscious referencing.

But when the meaning gradient is too strong, referencing becomes fixed to specific wells. This can appear as repetitive thought, compulsive attention, fixation, and traumatic re-referencing.

The meaning gradient directs consciousness, but an excessive gradient takes away the degrees of freedom of referencing.

15.6 Qualia and Measurement

This prediction concerns qualia and measurement.

Within the phenomenological boundary model, measurement is hypothesized to alter the boundary's retention conditions.

The proposed conversion into slow, reportable logs is expressed schematically, not as an established measurement law:

$$\text{measurement} \quad \Rightarrow \quad \text{boundary closure} \quad \Rightarrow \quad \text{sedimented residue} \tag{64}$$

This prediction is not the simple claim that qualia cannot be measured.

It distinguishes retained logs and reports from the live endpoint in the model; unchanged capture remains an unresolved hypothesis rather than a universal impossibility claim.

15.7 Measurement-Induced Residue Shift

Stronger measurement demand is hypothesized to favor stabilization as reportable residue and to alter live-state retention.

Increased precision demand may clarify sedimented residue; this directional proposal is not an established monotonic law or proof that qualia disappear.

This prediction requires treating measurement not as mere observation but as a regime-changing operation on the conscious state.

For comparable reported experiences, changes in reporting demand, options, verbalization, or numerical evaluation are proposed to alter the remaining logs. This comparison direction does not establish identity of the live experiences.

The model treats qualia reports as candidate residue shaped by measurement conditions, not as the live state itself. Comparing such residues remains a research direction; failure to capture a live endpoint unchanged would not by itself confirm the model.

15.8 Split-Brain Asymmetry

This prediction concerns split-brain experiments.

Behavioral/report asymmetry alone does not establish that consciousness doubles, nor that there is only one phenomenal subject.

The model predicts an asymmetry of log-reference accessibility as a candidate interpretation of pathway/report differences, not as an observed log mechanism.

Within the proposed access/report model, processing must connect to retained logs and a reporting route to count as explicitly accessible; this model condition is not a phenomenal criterion.

The background rationalization example is interpreted hypothetically as slow-log coherence completion. This offers a candidate account without deciding whether another phenomenal self is present.

15.9 Non-Local Regime Prediction

The temporal-regime hypothesis predicts that even matched local activity may be associated with different reports when temporal, semantic, and reporting conditions change. This is an untested comparison direction, not a demonstrated causal effect.

Correlation alone does not establish identity. The proposed comparison with temporal coupling is a separate hypothesis, not a consequence that validates this account.

This prediction does not deny local correspondence of neural activity.

Rather, it claims that the same site activity carries different conscious meaning depending on which temporal coupling it participates in within the consciousness window.

Accordingly, the model predicts differences in subjective report or behavioral response under these condition changes. An activity-matching tolerance, neural-to-log map, and executable protocol are not supplied; neither report nor behavior is identified with the phenomenal state.

15.10 Consciousness Window as Phase Diagram

The model organizes the qualitative predictions above into a finite parameter window and schematic phase diagram. This organization does not demonstrate that real conscious states occupy such a region or that altered states have already been classified. Whether the region can be empirically calibrated remains open.

$$0 < |\Delta v| < \Delta v_{\max}, \quad \mu_{\min} < \mu < \mu_{\max}, \quad \tau_{\min} < \tau < \tau_{\max}, \quad \mathcal{T}_{\min} < \mathcal{T}_{\Phi} < \mathcal{T}_{\max} \tag{65}$$

Within the schematic window, the model represents processing as neither fully synchronized nor fully separated, with stable log referencing as its regime label.

This window can be read as a phase diagram. Placing speed difference $|\Delta v|$ on the horizontal axis and coupling viscosity μ on the vertical axis, synchronization collapse, dissociation, fixation, fragmentation, and stable log referencing are positioned as distinct regimes.

Internal delay τ and meaning gradient $|\nabla\Phi|$ function as control variables that extend the state on this phase diagram in a depth direction.

The consciousness window is thus a multivariable formal region, labeled stable within the model, rather than an established empirical ON/OFF classifier.

Outside the window, the model proposes the following qualitative directions of change for future comparison:

- Collapse toward synchrony: automation, habituation, and reflexivity strengthen.
- Collapse toward separation: fragmentation, dream-like states, dissociation, and hallucinatory states strengthen.
- Too-low viscosity: fixation to meaning wells strengthens.
- Too-high viscosity: continuity of referencing is lost.
- Too-weak meaning gradient: referencing loses direction.
- Too-strong meaning gradient: referencing becomes fixed.

15.11 Testable Directions

This paper does not treat consciousness as a single directly measurable quantity.

However, directions for verifying conscious state transitions from changes in peripheral quantities are open.

Among verifiable directions, at minimum:

1. Correspondence between the time-scale difference of neural activity and the stability of subjective report.
2. Changes in τ through attentional manipulation.
3. Increase in referencing bias for semantically loaded stimuli.
4. Transformation of qualia-like reports by strengthened measurement and report demands.
5. Changes in speed difference, viscosity, and delay window in sleep, dreaming, anesthesia, and dissociative states.
6. Asymmetry of log-referencing pathways in linguistically non-reportable processing.
7. Clarification of residue and transformation of live qualia reports through measurement demand intensity.
8. Changes in conscious report due to differences in temporal coupling conditions for identical local activity.

9. Phase-diagram classification of conscious states on axes of speed difference and coupling viscosity.

None of these complete the framework of this paper.

Rather, they are entry points for investigating consciousness not as a product to be found but as a non-closure state across time scales.

15.12 Summary

The qualitative predictions of this paper reduce to one point:

The model labels membership in an intermediate effective region as its stable-consciousness window:

$$\text{stable consciousness} \quad \Leftrightarrow \quad (\Delta v, \mu, \tau, \mathcal{T}_\Phi, \nabla\Phi) \in \text{consciousness window} \tag{66}$$

In the biconditional above, “stable consciousness” is the model’s label for membership in the defined window, not a tested empirical or phenomenal iff. Operational calibration and an independent consciousness criterion are not supplied, and the classifier has not been tested. This window is not a fixed location.

It is the dynamic region in which temporal non-closure is maintained without rupture.

Consciousness is therefore not modeled here as a generated object.

The proposed reading associates opening and closing with these model conditions; the predictions remain qualitative directions for testing, not tested results.

16 Pathologies and Limits

This section organizes the proposed framework’s scope and qualitative regime comparisons, not its demonstrated clinical validity. The parameter directions below are model hypotheses, not measured assignments of the named states or established disease mechanisms.

”Pathologies” here does not mean medical diagnostic names.

This paper is not a theory for diagnosing psychiatric, neurological, or consciousness disorders.

What is treated here is structural deviation: the directions in which the log-referencing state can collapse when it departs from the consciousness window introduced in preceding sections.

The purpose of this section is therefore not to explain pathology but to clarify the limit conditions of conscious states.

16.1 Pathology as Regime Deviation

In the framework of this paper, stable conscious states are expressed as a window. Outside this window, consciousness does not merely ”weaken.” Collapse has direction:

- Direction of vanishing speed difference.

- Direction of excessive speed difference.
- Direction of too-low coupling viscosity.
- Direction of too-high coupling viscosity.
- Direction of too-weak meaning gradient.
- Direction of too-strong meaning gradient.
- Direction where live boundaries sediment excessively through measurement or report.

Each brings a different deformation of referencing state.

The crucial point is not linking these directly to disease names.

What this paper treats is not diagnostic classification but regime transition across time scales.

16.2 Collapse Toward Synchronization

The first deviation is the direction where speed difference becomes too small ($\Delta v \to 0$).

The boundary between fast and slow layers collapses into synchrony. Processing stabilizes and becomes efficient, but temporal thickness for referencing is lost.

In this direction, action automates, reactions become smooth, and explicit conscious referencing weakens.

This is not necessarily a bad state. Skilled movement, habituated action, and much reflexive processing are close to this direction.

But as one moves too far toward synchrony, reinterpretation of situations and re-referencing of logs becomes difficult. Consciousness does not so much stop as become unnecessary: processing runs, but the need for log referencing as opening diminishes.

16.3 Dissociation by Excessive Difference

The second deviation is the direction where speed difference becomes too large ($|\Delta v| \to \Delta v_{\max}$ or beyond).

The correspondence between fast and slow layers becomes unstable. Associations, predictions, images, and reactions run in the fast layer, but their stable sedimentation into the slow layer becomes difficult.

In this direction, the conscious state tends to fragment. Dreams, daydreaming, immersive states, dissociative states, and hallucinatory states can be described as part of this regime.

But care is needed here too. This paper does not equate these states. Increased speed difference and reduced log-reference stability are proposed as a heuristic structural comparison, not an established common dynamical structure of these distinct states.

This section therefore indicates not diagnostic classification but the direction of referencing-structure breakdown.

16.4 Fixation by Low Viscosity

The third deviation is where coupling viscosity μ is too low ($\mu \to 0$).

Fast and slow layers adhere excessively. Speed difference does not slip sufficiently; processing tends to bind strongly to specific logs or meaning wells.

In this direction, states such as repetitive thought, hyperfocus, fixation, and compulsive referencing tend to arise.

The crucial point is that what is occurring here is not merely "strength of meaning."

The model associates meaning-gradient/low-viscosity coupling with repeated attraction to the same well; this is a qualitative account of fixation, not a tested clinical mechanism.

Fixation is thus treated not as a content problem but as a viscosity problem of the referencing pathway.

Conscious content may feel vivid in this case, but vividness does not mean high degrees of freedom. Rather, fixation to meaning wells has strengthened and re-interpretability has decreased.

16.5 Fragmentation by High Viscosity

The fourth deviation is where coupling viscosity μ is too high ($\mu \to \infty$).

Fast and slow layers slip excessively against each other. Even though fast processing runs, it does not sufficiently sediment as a slow log.

Experience becomes difficult to retain; self-continuity and temporal coherence weaken.

In this direction, referencing fragmentation, sedimentation failure into memory, and reporting instability can occur.

This paper does not map this to a specific disease. What is stated here is only the structural tendency predicted when coupling viscosity is too high.

Processing exists, but does not stabilize as log referencing. This differs from collapse toward the synchrony side.

16.6 Weak Meaning Gradient

The fifth deviation is where the meaning gradient is too weak ($|\nabla\Phi| \to 0$).

Log referencing loses direction. Even though processing exists, which logs should be referenced and into which conceptual wells to fall becomes unclear.

In this direction, consciousness becomes diffuse, attention becomes scattered, and the outline of experience weakens.

But weak meaning gradient does not necessarily mean functional decline. Rest, diffuse thinking, free association, and creative wandering may utilize this region.

The problem arises when the meaning gradient is so weak that log referencing cannot form a stable terrain.

A weak meaning gradient can be soft exploration or loss of referencing — the difference is determined by the combination of speed difference, viscosity, delay, and gate state.

16.7 Excessive Meaning Gradient

The sixth deviation is where the meaning gradient is too strong ($|\nabla\Phi| \to \infty$).

Processing is strongly drawn into specific meaning wells. Log referencing becomes vivid, but degrees of freedom decrease.

In this direction, states such as emotionally strong memories, traumatic re-referencing, compulsive repetition, and excessive identification with narratives can be described.

The crucial point is that meaning gradient does not only strengthen consciousness.

Gradient directs referencing, but excessive gradient fixes referencing.

Strong meaning does not therefore necessarily mean deep understanding. An overly strong meaning field can rather take away the plasticity of understanding.

In this respect, $\mathcal{T}_\Phi$ is ambivalent: it is both the condition that opens log referencing and, in excess, the condition that fixes referencing.

16.8 Measurement-Induced Collapse

The seventh deviation models retention as shifting toward sedimented residue under measurement or report, following the hypothesis in §13.

The live boundary and stored or reported residue are distinct model categories; that distinction does not empirically establish unchanged-capture impossibility.

Stronger measurement demands are qualitatively predicted to favor the stabilization of verbalized, quantified, comparable logs. Their vividness is the vividness of residue, not identity with live qualia.

The possibility that observation or explanation demands change what is retained is a hypothesis for experimental design and introspection, not a monotonic law that every increase necessarily deforms its object.

Measurement and introspection are thus proposed as potentially regime-changing operations, leaving residue comparisons open. Failure to observe is not itself evidence for the model; correspondence with a live endpoint remains unresolved.

16.9 Mislocalization of Consciousness

The eighth deviation is treating consciousness as identical to local substrates or sedimented residue.

As in §14, neural/bodily dependence is a background premise. The non-local temporal-regime reading is a separate hypothesis, not a consequence established by that dependence.

What localization captures is local substrates, neural correlates, reportable logs, and post-measurement residue.

These are important. But identifying them with consciousness itself causes the live log-referencing regime to become invisible.

This error appears not only in discussions seeking the seat of consciousness but also in discussions seeking the self or qualia as specific objects within the brain.

This paper does not deny local substrates.

But local substrates can be necessary conditions for consciousness without being consciousness itself.

The model proposes associating conscious opening with a temporal, semantic, non-closed relation among candidate substrates; no calibrated carrier map or empirical sufficiency follows.

16.10 Limits of the Present Framework

The descriptions so far are intended to organize deviations of conscious states structurally.

But this paper has clear limits.

First, this paper is not a medical diagnostic theory. The regimes described here do not directly correspond to psychiatric or neurological classifications.

Second, the measurable definition of $\mathcal{T}_\Phi$ is not yet sufficiently established. The relation $\mathcal{T}_\Phi \sim \kappa \frac{\Delta v}{\mu} |\nabla \Phi|$ is at present a structural expression, not a concrete measurable quantity.

Third, how to map the local IFGT-inspired definitions of Φ, I, J_I, and the local flow $J_{\log}$ onto neural activity, linguistic report, and subjective report remains incomplete.

Fourth, treating qualia as boundary phenomena does not explain all of subjective experience. It repositions the qualia problem without eliminating subjectivity as a whole.

Fifth, this paper does not provide a unified equation computing consciousness. It chooses to retain temporal non-closure in its model description; this scope does not establish uncomputability or the impossibility of a closed mathematical account.

The theoretical scope of this paper lies therefore not in the complete description of consciousness itself but in the specification of the structural conditions under which consciousness can obtain.

16.11 Why the Limits Matter

These limits are both weaknesses of the paper and safety mechanisms of the theory.

Attempting to measure consciousness as a single variable immediately returns the theory to the generation model.

Treating qualia as stored objects loses their character as boundary phenomena.

Linking pathologies directly to diagnostic names loses the flexibility of structural regimes.

Placing consciousness at local sites makes the live log-referencing loop invisible.

This paper therefore deliberately leaves several points unclosed.

Not closing is not vagueness.

It is because the object this paper treats — consciousness — is itself a temporal structure maintained in non-closure.

If the theory rushes to complete closure, it loses the consciousness that is its object.

Making this constraint explicit is the purpose of this section.

16.12 Summary

Pathologies in this paper are not disease classifications but directions of deviation from the consciousness window.

- $\Delta v \to 0$: synchronization, automation, weakening of conscious referencing.
- Excessive $|\Delta v|$: fragmentation, dream-like states, dissociation-like states, hallucinatory-like states.
- $\mu \to 0$: fixation, repetition, fixing to meaning wells through low viscosity.
- $\mu \to \infty$: sedimentation failure, fragmentation of referencing through high viscosity.
- $|\nabla \Phi| \to 0$: disappearance of meaning gradient, weakening of outline.
- Excessive $|\nabla \Phi|$: over-fixation to meaning wells, decrease of plasticity.
- Excessive measurement demands: hypothesized shift of live-state retention toward sedimented residue.
- Excessive localization: mistaking local substrates or residue for consciousness itself.

None of these determine the presence or absence of consciousness in binary terms.

They are structural classifications showing in which direction the conscious state deforms and under which conditions it leaves the stability window.

17 Relation to the Note

This section organizes the relation between this paper and the companion note series.

This paper is not a replacement for the note series.

The note series is an intuitive reading on consciousness, beginning from experience, metaphor, unease, and everyday sensation.

This paper, by contrast, is the theoretical foundation for fixing the structure behind that.

The role of this paper is therefore not to overwrite the note series with theory but to reposition what was intuitively shown in the note series within the structure of temporal non-closure, speed-scale difference, log referencing, and meaning gradient.

17.1 The Note as Intuitive Layer

In the note series, what readers should first encounter is not equations.

What readers should first encounter is a sense of unease about their own conscious experience.

Why does the body begin moving before one thinks one has decided? Why does a sensation change slightly the moment it is put into words? Why do strong memories and emotions repeatedly return to consciousness? Why does consciousness feel as though it is inside oneself, yet cannot be found when searched for?

The note series handles these questions intuitively.

Theoretical terms can be minimal there.

What matters is that readers feel: "Perhaps consciousness is less something generated and more something that opens."

17.2 The Present Paper as Structural Layer

This paper, on the other hand, structures the intuitions presented in the note series.

The core opening and window statements below retain the model-local scope of §§1 and 7. They distinguish a structural definition from the proposed conscious-state identification; their necessity is architectural, not an empirical or universal criterion:

- Consciousness is not a product but a log-referencing state.
- Consciousness is not a single site but a temporal non-closure regime.
- Consciousness depends on speed difference Δv between fast and slow layers.
- Consciousness requires coupling viscosity μ and internal delay τ.
- Log referencing is directed by meaning gradient $\nabla\Phi$.
- Effective torque-like quantity $\mathcal{T}_\Phi$ represents the coupling of speed difference and meaning gradient.
- Qualia name the live sedimentation frontier in the phenomenological model; their identification with experience is proposed.
- Measurement is hypothesized to shift retention of that live boundary toward residue.
- The model proposes a non-local temporal regime among candidate substrates; neural dependence alone does not establish it.

These need not be brought to the foreground in the note series.

But they must be retained as the structure behind the note series.

17.3 What Should Flow Back to the Note

What should flow back from this paper to the note series is not all the equations.

What should flow back is the minimal structure that helps readers understand.

Concretely:

1. The expression that consciousness "opens" rather than "is produced."
2. The intuition that consciousness opens between fast processing and slow memory sedimentation.
3. The organization of the ego not as subject but as transformation point of speed difference.

4. The organization of qualia not as stored objects but as live foreground at the sedimentation boundary.

5. The hypothesis that measurement and introspection alter retention toward logs, distinguishing live qualia from residue without proving unchanged-capture impossibility.

6. The proposed organization of consciousness through temporal conditions rather than a single brain-point identity.

These are sufficiently intuitive for note series readers while also connecting to the theoretical foundation of this paper.

17.4 What Should Not Flow Back Directly

Conversely, some things should not be brought directly from this paper into the note series.

First, equations should not be brought in excessively. Δv, μ, τ, $\mathcal{T}_\Phi$, $\nabla\Phi$ are important structural variables in this paper, but foregrounding all of them in the note series breaks the flow as a reading.

Second, pathological deviations should not be treated as diagnoses. The pathologies of this paper are directions of deviation from the consciousness window, not psychiatric classifications. Even when treating these in the note series, they must be handled carefully as deformations of conscious states, not medical explanations.

Third, IFGT/VED terminology should not be brought too prominently to the foreground. The role of the note series is not to make readers understand theory names but to let readers experience structure. Theory should support from behind, not advance to the front.

17.5 Layered Translation

The relation between this paper and the note series is not simple summarization.

It is translation across layers.

At L3, the following schematic expression records the proposed association, not an established phenomenal identity:

$$\text{Consciousness} \sim \text{Log Referencing}(\Delta v, \mu, \tau, \mathcal{T}_\Phi, \nabla\Phi) \tag{67}$$

But in the note series, there is no need to present this directly.

In the note series, it can be restated as:

Consciousness is the passage that temporarily opens between fast-flowing processing and slowly sinking memory.

This pedagogical restatement presents the model at different explanatory resolutions. It is not a typed cross-domain translation or evidence that distinct source windows have one structure.

This is the relation between the note series and this paper.

17.6 Relation to Appendices

Between the note series and this paper, supplementary materials and intermediate explanatory layers are placed.

This paper fixes structure as L3.

Supplementary materials explain that structure to readers as L2.

Where necessary, L2.5 serves as a bridge that returns the meaning of equations to intuition.

The note series is the experiential layer readers encounter first, as L1.

Not conflating these layers is important.

L3 must not dominate L1.

L1 must not blur L3.

Each layer has a different role.

The purpose of this paper is not to harden the text of L1 but to fix the structure behind it so that L1 can write with freedom and ease.

17.7 Summary

The relation between this paper and the note series is repositioning, not replacement.

- The note series is the layer of experience and intuition.
- This paper is the layer of structure and equations.
- Not everything in this paper needs to be brought into the note series.
- What should flow back to the note series is the minimal structure that helps readers' intuition.
- Theoretical terms and equations support through supplementary materials and intermediate layers when needed.
- L3 does not overwrite L1.
- L3 supports L1 from behind.

This paper is therefore not the completed version of the note series.

This paper is the skeleton behind the note series — enabling the note series to breathe freely as a reading.

18 Discussion

The core opening and window descriptions in this section retain the model-local scope of §§1 and 7: structural definition and proposed conscious-state interpretation are distinct, and necessity refers to this architecture.

This section organizes the positioning of this paper, differences from existing theories, theoretical implications, and future tasks.

The central claim of this paper is that consciousness should not be treated as a product.

Consciousness is not modeled here as an object produced somewhere.

Consciousness is a log-referencing state that opens when a certain speed difference, coupling viscosity, internal delay, and meaning gradient obtain between fast predictive processes and slow log-sedimentation processes.

This view does not erase consciousness.

Rather, it removes consciousness from its conventional positions — as source of causation, object within the brain, content of representation, container of qualia — and repositions it as a temporal non-closure regime.

18.1 Relation to Global Workspace-Type Models

The following Global Workspace and IIT comparisons locate this model conceptually. They claim neither refutation nor empirical superiority, and the shared vocabulary supplies no carrier identity.

In Global Workspace-type theories, consciousness is often understood as information becoming globally available, or being broadcast to multiple processing systems.

This paper can connect with this orientation in part.

In particular, the point that conscious states obtain not at a single site but as a relation among multiple processing systems has affinity with the idea of global availability.

But this paper does not define consciousness as ”broadcasting” or ”sharing” itself.

What this paper emphasizes, before information is widely used, is in which time scale it becomes referenceable.

Fast-layer processing sediments into the slow layer and becomes referenceable as a log.

At that point, the conscious state opens.

Spatial breadth is therefore less important in this paper than temporal transformation.

Information becoming broadly available may be part of the result or conditions of conscious state, but it is not consciousness itself.

18.2 Relation to IIT-Type Models

In IIT-type theories, consciousness is associated with the degree of information integration or structural indivisibility.

This paper also does not reduce consciousness to simple local processing, and emphasizes that multiple elements form a relational structure.

In that sense, treating consciousness as structural relation rather than isolated parts has some common ground.

But this paper does not define consciousness as an integration quantity.

Consciousness is treated not as the magnitude of integrated information but as the temporal condition under which log referencing opens.

If integration is too strong, structure becomes fixed and may lose plasticity.

If integration is too weak, processing dissipates and log referencing cannot obtain.

In this paper, therefore, the intermediate state that does not reach complete closure is emphasized over integration itself.

Consciousness is not maximum integration but a non-closure stability window.

18.3 Relation to Higher-Order and Self-Model Theories

In Higher-Order and self-model theories, consciousness is explained as one representation capturing one's own state, or as a self-model representing a state.

This paper acknowledges the importance of self-reference and re-referencing.

Indeed, for log referencing to obtain, current processing must have a relation to slow-layer logs, and that relation must in turn influence current processing.

In this sense, this paper does not deny self-reference.

But this paper does not hold that self-representation generates consciousness.

Within this model, the self is represented as maintained slow-layer terrain and the ego as a conversion role, not as hidden causes of consciousness. These roles do not establish a phenomenal subject or exclude higher-order accounts empirically.

The ego is not a subject but the position at which speed difference is transformed.

Self-models are important, but they are terrain formed within the log-referencing loop, not the source of consciousness generation.

18.4 Relation to Predictive Processing and Free Energy Approaches

This paper emphasizes prediction, error, and forward models in the fast layer.

In this respect, it can connect with predictive processing and the free energy principle.

The fast layer anticipates the future, generating action candidates and error corrections.

The slow layer sediments part of this and changes the terrain for subsequent prediction.

This loop contains the structure of prediction and updating.

But the focus of this paper is not the accuracy of prediction or minimization of free energy itself.

What matters is how prediction sediments into a log-referenceable structure.

Even with fast prediction, if it does not reach the slow layer, the conscious state does not open.

Conversely, if sedimentation is too strong, prediction becomes fixed to specific meaning wells.

This paper therefore does not view predictive processing as a sufficient condition for consciousness.

Within this model, prediction is hypothesized as a necessary functional component together with speed difference, viscosity, delay, meaning gradient, and non-closure. This comparative suggestion establishes neither universal necessity nor the sufficiency of predictive processing for consciousness.

18.5 Why the Model Is Not Reductionist

This paper does not treat consciousness as a mysterious entity.

But this does not mean reducing consciousness to simple physical quantities or neural activity.

Rather, to avoid simple reduction, this paper treats consciousness as a state domain.

Consciousness is not I itself, not J_I, not $J_{\log}$, not Φ, not any one of Δv, μ, τ, $\mathcal{T}_\Phi$.

Consciousness is the regime that opens when these enter a certain relation.

In this sense, the position of this paper is neither eliminativist nor dualist.

It does not add consciousness as a special entity.

But it does not collapse consciousness into a single physical quantity either.

It treats consciousness as the relation among multiple layers.

18.6 Why the Model Is Not Anti-Neuroscience

This paper proposes a temporal regime rather than a single-site consciousness identity; non-localization is a model reading, not an empirically established exclusion.

But this is not a negation of neuroscience.

Rather, it is a framework for interpreting neuroscientific findings more carefully.

Certain brain regions, circuits, signals, oscillations, and activity patterns can certainly correlate with conscious states.

These are important local substrates.

But those local substrates cannot be identified with consciousness itself.

In this account, log-loop participation is a candidate interpretation of neural activity. Neither the background neural-dependence premise nor the correlation/nonidentity distinction establishes that map.

Which activity corresponds to the fast layer? Which activity corresponds to sedimentation into the slow layer? Which pathway supports reportability? Which activity forms the meaning gradient?

Asking in this way repositions neuroscientific data not as searching for the seat of consciousness but as investigating the conditions under which conscious states open.

18.7 The Role of IFGT and VED

In this paper, IFGT and VED have been used not as completed theories that fully explain consciousness but as a coordinate framework for repositioning consciousness.

IFGT provides the structure by which log referencing is directed along the meaning gradient, through projected uses of information density I, information flow J_I, local log-referencing flow $J_{\log}$, and information potential Φ.

VED provides the structure of difference, flow, and persistence in which complete closure is not realized as an ordinary state.

Connecting these two, consciousness is positioned as follows:

Consciousness is modeled as the state in which log referencing directed by the projected information potential is open within a temporal non-closure regime that reaches neither complete synchrony nor complete separation.

This formulation does not reduce consciousness to a physical field.

It is a coordinate system for fixing the variables and relations required when speaking of consciousness.

The intelligence paper [1] also adopts projected uses. Its discussion of crossing beyond the individual and this paper's intra-individual temporal non-closure are compared as different architectural windows. Complementarity is a heuristic reading, not a common non-closure law or a typed translation. Downstream reuse does not supply evidence for this paper's conscious-regime interpretation.

18.8 Theoretical Implications

The theoretical implications of this paper include several points.

First, the question in consciousness research shifts from "where is consciousness generated?" to "under which conditions does log referencing open?"

Second, the live sedimentation frontier remains a phenomenological definition of qualia, while identification with experience and its explanation remain distinct, proposed readings.

Third, measurement is hypothesized to shift retention toward sedimented residue. The live/residue distinction does not establish universal capture impossibility; comparisons remain possible and the live endpoint remains unresolved.

Fourth, neural correlates are distinguished from an established consciousness identity; their relation to the log-referencing loop remains a candidate interpretation.

Fifth, pathologies and altered states can be structurally organized not as disease classifications but as directions of deviation from the consciousness window.

These do not negate consciousness research.

Rather, they are translations that move existing data and discussions from the generation model to the opening model.

18.9 Remaining Questions

Unresolved tasks remain in this paper.

First, the measurable definition of $\mathcal{T}_\Phi$ is incomplete. How to map speed difference, coupling viscosity, and meaning gradient onto neural activity, bodily states, subjective reports, and linguistic logs is a future task.

Second, how to map IFGT's I, J_I, Φ and the local flow $J_{\log}$ onto brain data and behavioral data must be further refined.

Third, how to empirically estimate the consciousness window remains undetermined.

Fourth, methodological ingenuity is needed for how to experimentally handle qualia sedimentation residue.

Fifth, how the self, language, and social interaction extend the log-referencing structure exceeds the scope of this paper.

These tasks do not negate this paper.

The paper's openness to further work may be compared methodologically with its non-closure vocabulary. An unfinished theory is not evidence that consciousness itself is non-closed.

18.10 Toward Consciousness, Self, and Intelligence

This paper treated consciousness as the opening of log referencing.

The account suggests a conceptual handoff beyond consciousness. The following order is explanatory and architectural, not a proof of necessary generation or a universal developmental sequence.

The self can be understood as the stable terrain formed in the slow layer through the repetition of openings.

Intelligence can be understood as the structure that uses that terrain to extend future prediction, language, social logs, and interaction with the external environment.

In this sense, this paper is not the endpoint of consciousness theory but the bridge to self theory and intelligence theory.

Consciousness is not modeled as a generated object. Consciousness opens.

The self sediments through the repetition of that opening.

Intelligence flows back into the world again using that sedimented terrain.

Following this flow is the next task.

Position within the Four-Layer Comparison The proposed consciousness/self/intelligence order is a conceptual handoff within the physical, biological, consciousness, and intelligence comparison [1]. Bio's quasi-closure vocabulary supplies structural motivation, not a derivation or dedicated fast/slow validation. The intra-individual precursor is architectural, not proof of Intelligence's crossing necessity. The four layers are compared heuristically through "trying to close but not closing completely," not identified as one generative grammar. Their bridges remain open.

18.11 Summary

This section organized the positioning of this paper.

- This paper treats consciousness not as a product but as the opening of log referencing.
- It has points of contact with Global Workspace, IIT, Higher-Order, and Predictive Processing discussions, but is not identical to them.
- This paper does not reduce consciousness to a single site, single quantity, or single representation.
- It does not deny neuroscience; it proposes neural/log-loop correspondences without treating correlation as identity or model validation.

- IFGT and VED are used not as theories that fully explain consciousness but as coordinate systems for positioning it.
- The claim of this paper is shifting the question in consciousness research from "generation" to "opening conditions."
- Remaining questions keep the theory open to research; the resemblance to non-closure is methodological, not evidence about the object.

The significance of this paper therefore does not lie in explaining consciousness as a closed object.

It lies in repositioning consciousness — as a log-referencing state that opens within temporal non-closure — in a form that allows it to be questioned again.

19 Conclusion

This paper has reformulated consciousness not as an object generated internally but as a log-referencing state that opens between processes at different time scales.

The central proposal separates the model definition of a log-referenceable relation from its interpretive association with consciousness. Neither the definition nor the window establishes phenomenal identity, empirical classification, or demonstrated consciousness maintenance.

Consciousness is not modeled here as a generated object.

Consciousness opens when fast predictive processes and slow log-sedimentation processes couple within a range that reaches neither complete synchrony nor complete separation.

This opening involves speed difference Δv, coupling viscosity μ, internal delay τ, meaning gradient $\nabla\Phi$, and effective torque-like quantity $\mathcal{T}_\Phi$.

But none of these individually is consciousness itself.

Consciousness is the temporal non-closure regime that obtains when these enter a certain relation.

19.1 What Has Been Proposed

This paper first introduced SSO as a proposed temporal architecture, treated as necessary within the model but not sufficient for consciousness.

SSO is a structure in which processing layers running at different speeds are coordinated without central control.

The model assigns the fast layer anticipation of the future and the slow layer sedimentation of the past.

A finite speed difference remaining between them opens a referencing relation between current processing and past logs.

Next, Torque Converter Attention was introduced.

This retains the torque-converter analogy as an effective temporal transformation model of attention, not an established neural mechanism.

In this position, the ego is understood not as a subject but as the position at which the speed difference between fast and slow layers is transformed.

Furthermore, using IFGT and VED, log referencing was repositioned within meaning gradient and non-closure dynamics.

IFGT describes which logs are easily referenced through projected uses of information density I, information flow J_I, local log-referencing flow $J_{\log}$, and information potential Φ.

VED provides the structure of difference and flow in which complete closure is not realized as an ordinary state.

Connecting these two, consciousness was described as the state in which log referencing directed by the meaning gradient is open within a temporal non-closure regime.

19.2 What Has Been Repositioned

This paper repositioned several important concepts.

First, consciousness was repositioned not as an object but as a state domain.

Consciousness is not an object stored somewhere, nor an output generated within the brain.

It is a regime that opens when processing and logs enter a referenceable relation.

Second, qualia were located at the live sedimentation frontier as a phenomenological definition. The identification with experience is proposed, not established, and naming this position does not solve the hard problem.

Qualia are not denied.

But they are not stored objects.

What is stored is not qualia themselves but the log structure remaining after qualia have passed through.

Third, measurement was proposed as an operation that may shift live-state retention toward sedimented residue, not as a validated transformation mechanism.

Measurement is not meaningless.

The model distinguishes retained traces and reports from the live state itself. This does not prove unchanged-capture impossibility, and neither noncapture nor report alone supplies an independent phenomenal criterion.

Fourth, non-localization was repositioned not as mystification but as a consequence of treating consciousness as a temporal relation.

Neural/bodily dependence is retained as a background premise. The temporal, semantic, non-closed relation among candidate local substrates is separately proposed as a conscious regime. Neither a carrier map nor phenomenal sufficiency is established by the dependence premise.

Fifth, named pathologies and altered states were retained as heuristic comparisons and research directions, not diagnoses or measured members of a common dynamical regime.

Through this organization, the conscious state is treated not as a binary presence or absence but as a dynamic regime with a direction of collapse.

19.3 The Central Formula

The central structure of this paper can be summarized in the following expression:

$$\text{Consciousness} \sim \text{Log Referencing}(\Delta v, \mu, \tau, \mathfrak{T}_\Phi, \nabla\Phi) \tag{68}$$

This expression does not compute consciousness.

It is the coordinate showing the structural variables required when speaking of consciousness.

More concretely, the following inequalities define the model window; the symbolic bounds remain uncalibrated, not measured consciousness thresholds:

$$0 < |\Delta v| < \Delta v_{\max} \tag{69}$$

$$\mu_{\min} < \mu < \mu_{\max} \tag{70}$$

$$\tau_{\min} < \tau < \tau_{\max} \tag{71}$$

$$\mathfrak{T}_{\min} < \mathfrak{T}_\Phi < \mathfrak{T}_{\max} \tag{72}$$

Within this model window, processing is represented as neither fully synchronized nor fully separated, and stable referencing is the proposed regime.

Outside the window, the model organizes proposed directions of change; classifying real conscious states remains an empirical task.

Consciousness is therefore not a point but a window.

Not an object but a regime.

Not a product but an opening.

19.4 Why the Model Remains Open

This paper does not provide a complete unified equation of consciousness. This is a scope statement, not a theorem of uncomputability or a prohibition on mathematical work.

This is not an omission but an intentional constraint.

A closed-object description would set aside the non-closure emphasized in this model; this is a choice of representation, not proof that a closed mathematical account is impossible.

Treating qualia as stored objects loses their character as boundary phenomena.

Identifying neural correlates with consciousness itself makes the structure as temporal regime invisible.

Linking pathologies directly to diagnostic names loses the flexible structure of directions of deviation from the consciousness window.

This paper therefore deliberately does not close completely.

Not closing completely is not abandonment of explanation.

The non-closure vocabulary offers a methodological analogy for keeping the research program open, not evidence about consciousness drawn from the theory's incompleteness.

Theory and its proposed object can be compared in this methodological sense only.

If consciousness is non-closed, consciousness theory too must not rush to complete closure.

19.5 Toward Further Work

Future tasks are clear.

First, $\mathcal{T}_\Phi$ must be brought closer to concretely measurable quantities connected to neural activity, bodily states, subjective reports, and linguistic logs.

Second, methods for mapping IFGT's I, J_I, Φ and the local flow $J_{\log}$ onto brain data and behavioral data must be refined.

Third, the consciousness window must be examined against states such as sleep, dreaming, anesthesia, immersion, dissociation, skilled movement, and verbal report.

Fourth, experimental designs for handling qualia sedimentation residue must be developed.

Fifth, how the self, language, and social interaction extend the log-referencing structure must be addressed.

This last task is a conceptual handoff from the consciousness paper to self and intelligence research. The following sequence is an architectural proposal, not a generation proof or a transfer of empirical authority from downstream work.

Consciousness is the opening of log referencing.

The self is modeled as terrain maintained through repeated openings, not a phenomenal subject generated from nothing. Experienced continuity is a proposed correspondence; first-log initialization and log ownership remain open.

Intelligence is the motion that uses that sedimented terrain to extend into future prediction, language, social logs, and interaction with the external environment.

19.6 Final Statement

Consciousness is not produced somewhere.

Consciousness is the passage temporarily opened between fast-flowing processing and slowly sinking logs.

There is speed difference.

There is delay.

There is viscosity.

There is meaning gradient.

And there is flow that does not close completely.

While this passage is open, processing does not merely flow — it references past logs, updates the terrain of the self, and reconstructs the relation with the world.

This is the structural state the proposal associates with consciousness.

The conclusion of this paper therefore returns to the proposition with which it began.

Consciousness is not modeled here as a generated object.

The model proposes that consciousness is associated with log referencing within temporal non-closure; it does not turn that association into an established phenomenal or empirical identity.

20 Correspondence Map: This Paper ↔ Conceptual Topography

For readers who will also read Conceptual Topography (CT) alongside this paper, the correspondence between the main concepts of both papers is shown here.

This table proposes non-reductive comparisons between this paper and CT. Neither paper grounds or validates the other. Conscious opening and conceptual terrain formation are compared through heuristic motifs of differentiation and stabilization, not as a common VED×IFGT generative law. The projected-uses stance in §6.1.1 applies to every row: symbols retain their source-local domains, and no typed translation, common units, preserved invariant, or shared carrier is supplied.

This paper	CT (Conceptual Topography)	Proposed role comparisons
Slow-layer sedimentation structure	Information potential Φ landscape	Sedimentation compared with CT's Φ^* terrain; no shared limit claimed
SSO (temporal non-closure)	Invisible GPU (landscape operating below consciousness)	Comparison of indirect landscape influence; Φ remains source-local
TCA (transformation of speed difference)	Token coupling $\lambda \sum_i \ell_i \cdot \nabla\Phi$	Comparable temporal/spatial transformation roles, not isomorphism
Log-referencing direction by meaning gradient $\nabla\Phi$	Concept definition: locally stable region $\nabla\Phi \approx 0$	Analogous directional role of $\nabla\Phi$, not the same quantity
Log referencing (state where consciousness opens)	Traversal cost reduction (state where understanding obtains)	Referencing/traversal compared using source-local $\nabla\Phi$ roles
Qualia (live foreground at sedimentation boundary)	Insight (rapid topological transition)	Boundary-like transitions compared across domains; no phenomenal identity
$I(x,\tau) > 0$ (exclusion of complete closure)	$I(x,\tau) > 0$ (exclusion of complete conceptual closure)	Positivity marks the exclusion of ordinary complete closure in each window; it does not by itself identify consciousness with the conceptual landscape

Reading suggestions:

For readers who read this paper first: compare the proposed temporal conditions of conscious opening with CT's conceptual terrain formation. The geometry vocabulary offers

an analogy for referencing bias, not an established determination of conscious content by a CT field.

For readers who read CT first: SSO offers a comparative temporal architecture for discussing referenceability. This reading does not make SSO a derived or independently established necessary condition of CT terrain formation. The qualia/insight row is likewise an analogy, not CT support for a phenomenal criterion or the identity of two experiences.

Mutual reading extends the comparison to the VED physical layer, the Bio-IFGT biological layer, and the intelligence layer [1]. Four projected windows can be compared through recurring non-closure and differentiation motifs without reduction, mathematical isomorphism, or a shared generative mechanism. The Consciousness-to-self-to-Intelligence sequence is a conceptual handoff and explanatory order, not a universal developmental sequence or generation proof. Downstream reuse does not validate this paper; upstream closure supplies no missing bridge.

References

[1] Yuhki Endoh. Structural constraints on intelligence as a dynamical phenomenon, 2026. Intelligence Section, Part I.